



Wheel-Rail Contact: an introduction

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Overview

- 1 Getting “Inside” the Wheel-Rail Interface

- 2 Fundamental #1: Contact Pressure & Geometry

- 3 Fundamental #2: Traction & Creepage

- 4 Reality Check: The Third Body Layer

- 5 Steering Forces: A Practical Illustration

But first... a question to consider:

We often hear statements that sound something like:

1. “The outer (high) rail in curve is longer than the inner (low) rail.”
2. “Railway wheels are coned (tapered), so the wheel radius in contact with each rail changes when the wheelset shifts laterally.”
3. “The difference in rolling radius between the left and right wheels *allows* the wheelset to steer around the curve by compensating for the different lengths of each rail.”

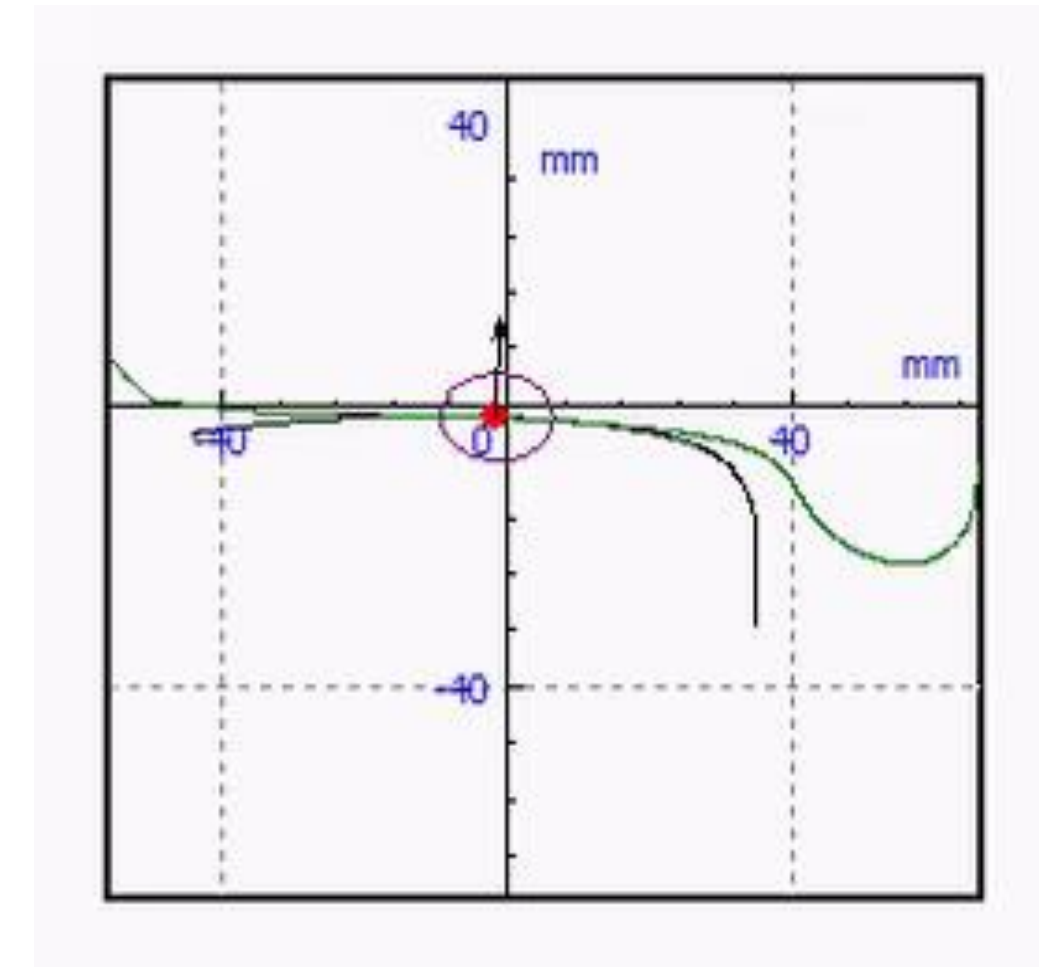
This is a nicely intuitive description, but...





If that were the complete story...

- As a wheel “climbs” the rail, its radius starts getting much larger, very quickly.
- Why wouldn’t the resulting rolling radius difference between the wheels be enough to steer the wheelset back on track instead of continuing to derail?

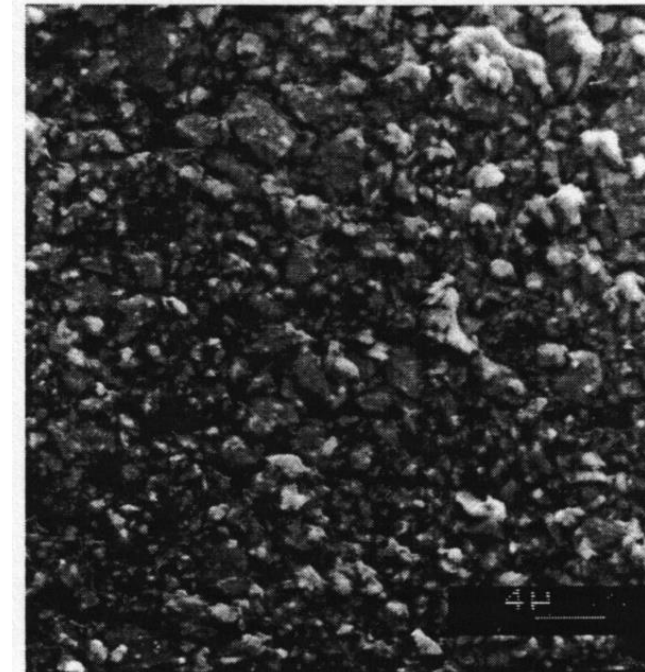
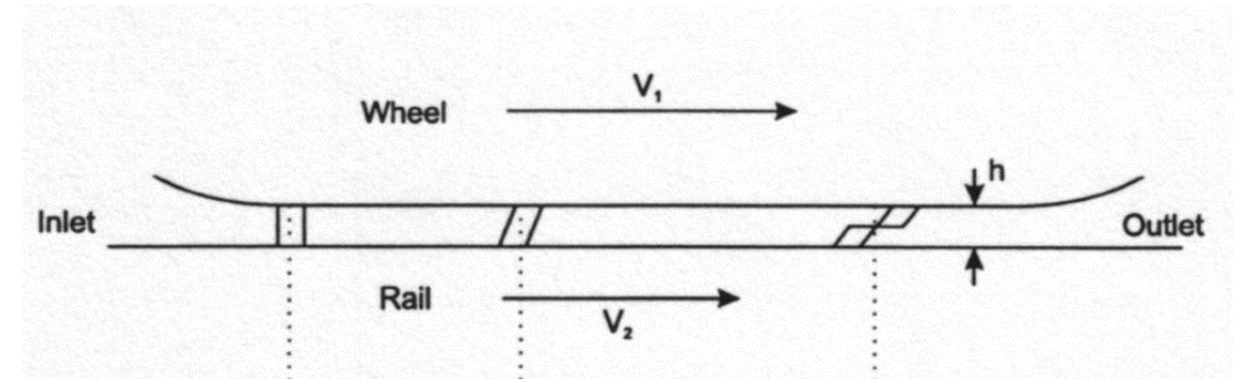
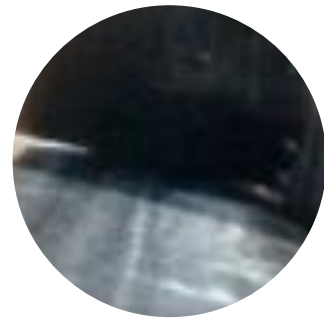


Getting “Inside” the Wheel-Rail Interface

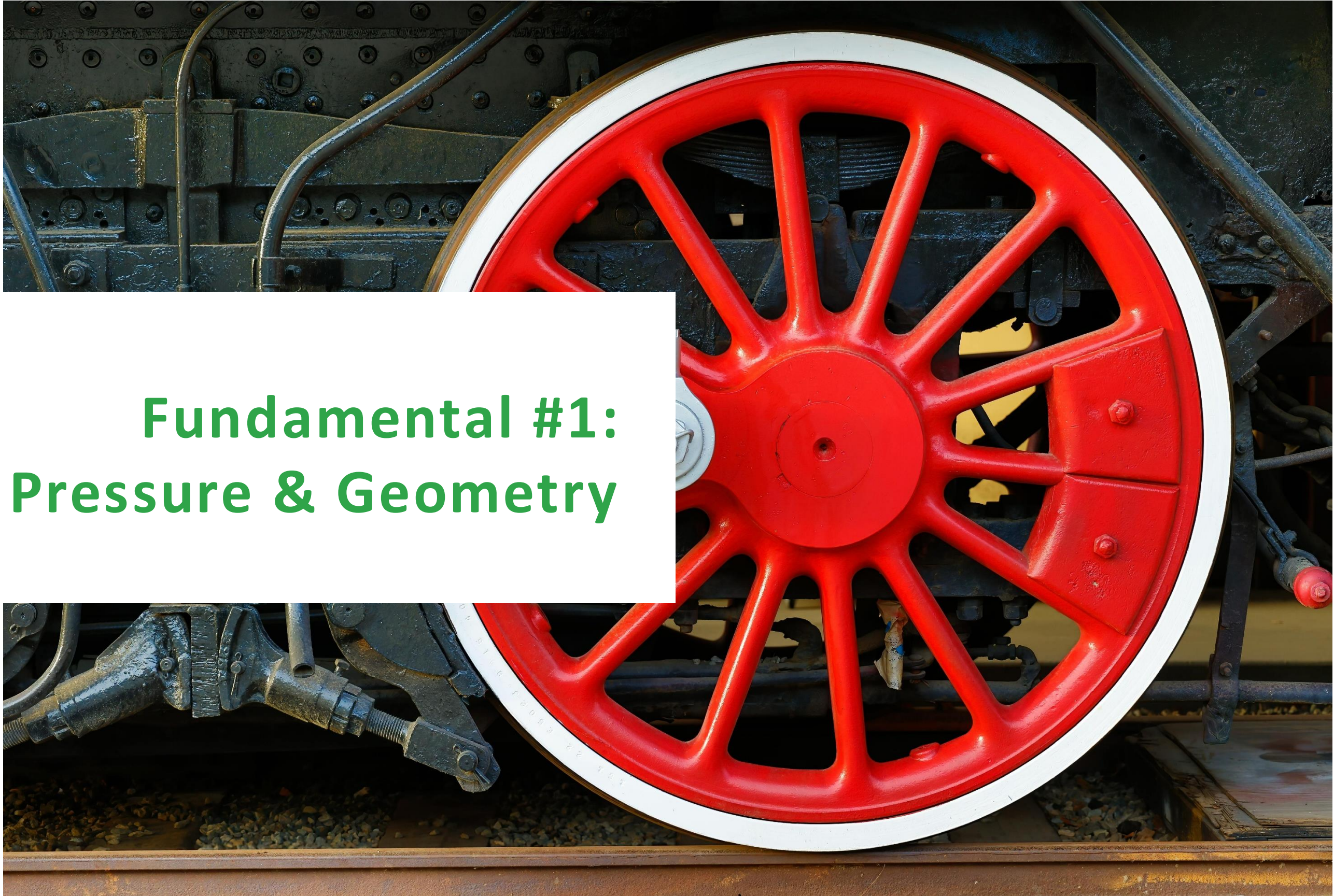




Getting “Inside” the Wheel Rail Interface



Kalousek, J, K Hou, E Magel, and K Chiddick. “The Benefits of Friction Management: A Third Body Approach.” In *Proceedings of the World Congress on Railway Research (WCRR 1996)*, 8. Colorado Springs, 1996.

A photograph of a red steam locomotive wheel on a track. The wheel has a white rim and is surrounded by dark, industrial machinery. A white rectangular text box is overlaid on the left side of the wheel.

Fundamental #1: Contact Pressure & Geometry



The “contact patch”

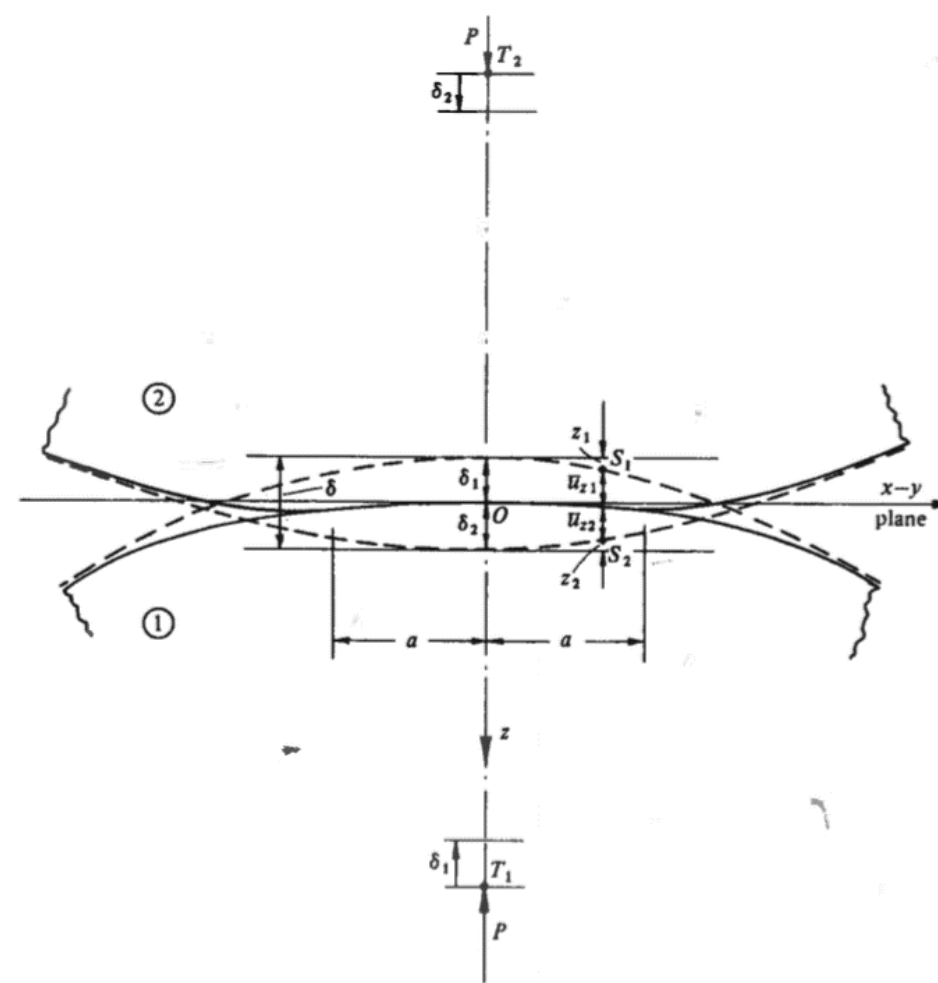
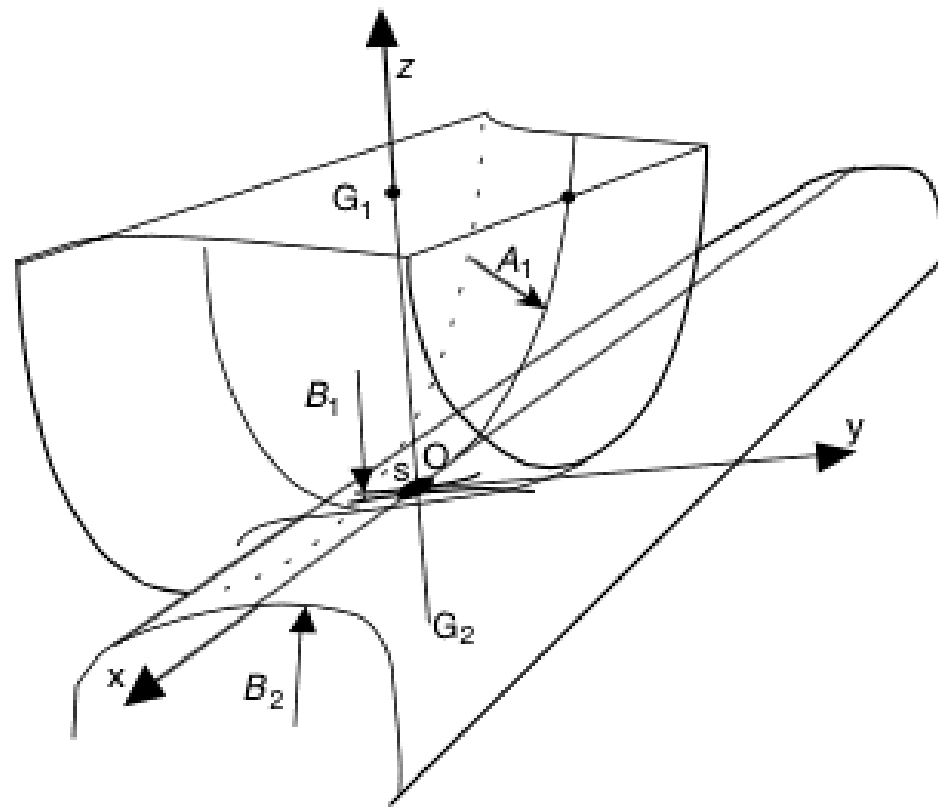
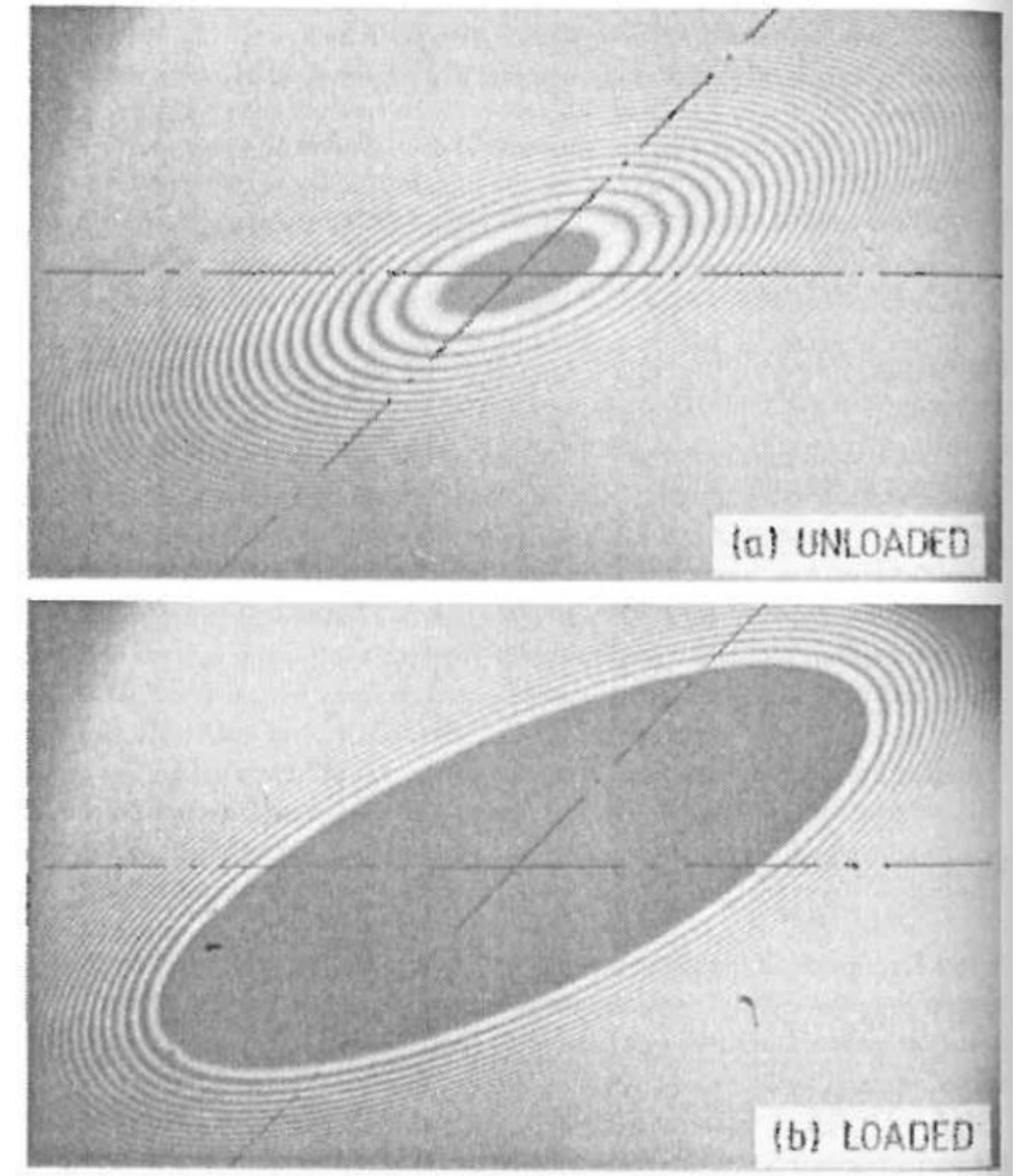


Fig. 4.1. Interference fringes at the contact of two equal cylindrical lenses with their axes inclined at 45° : (a) unloaded, (b) loaded.



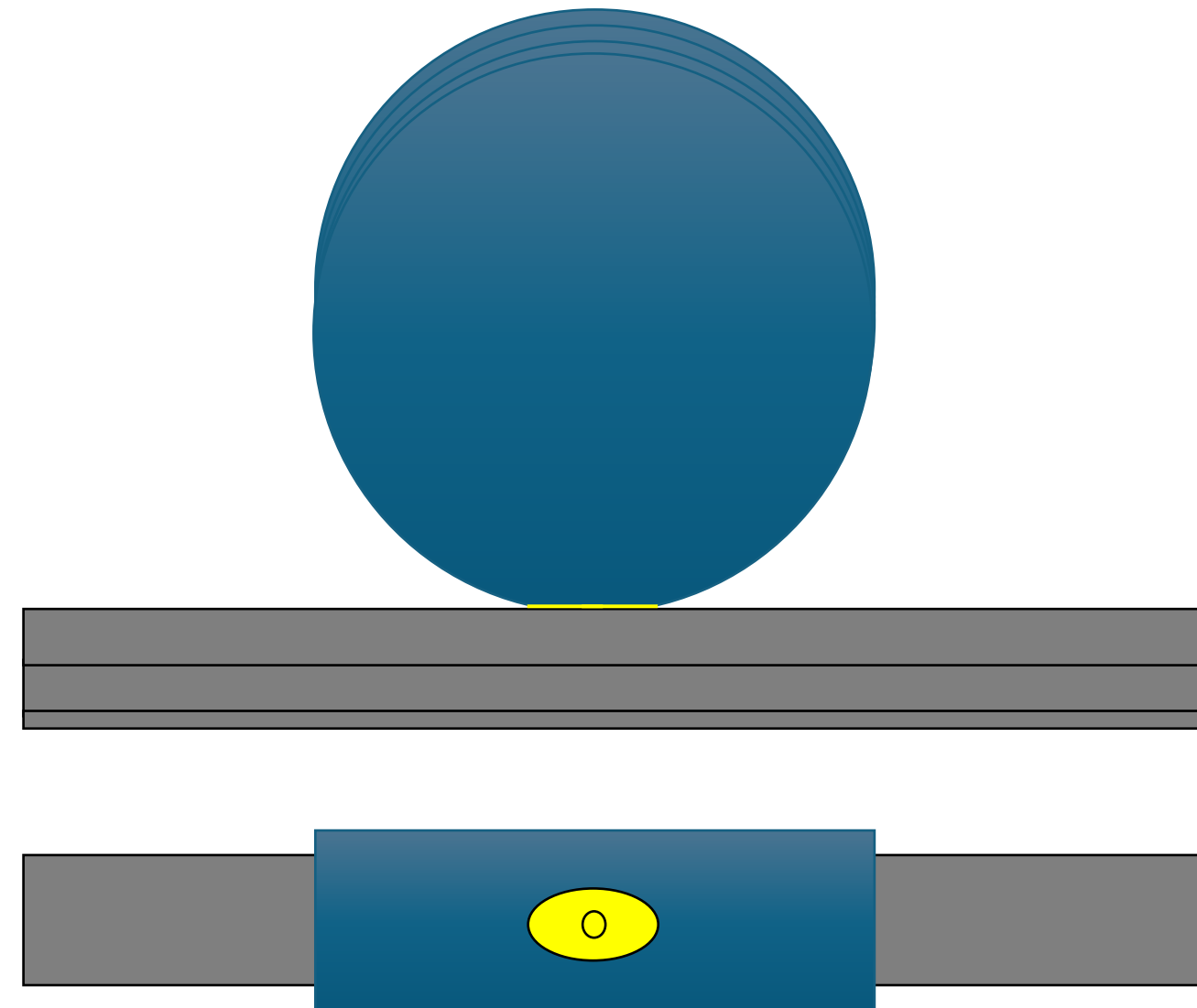
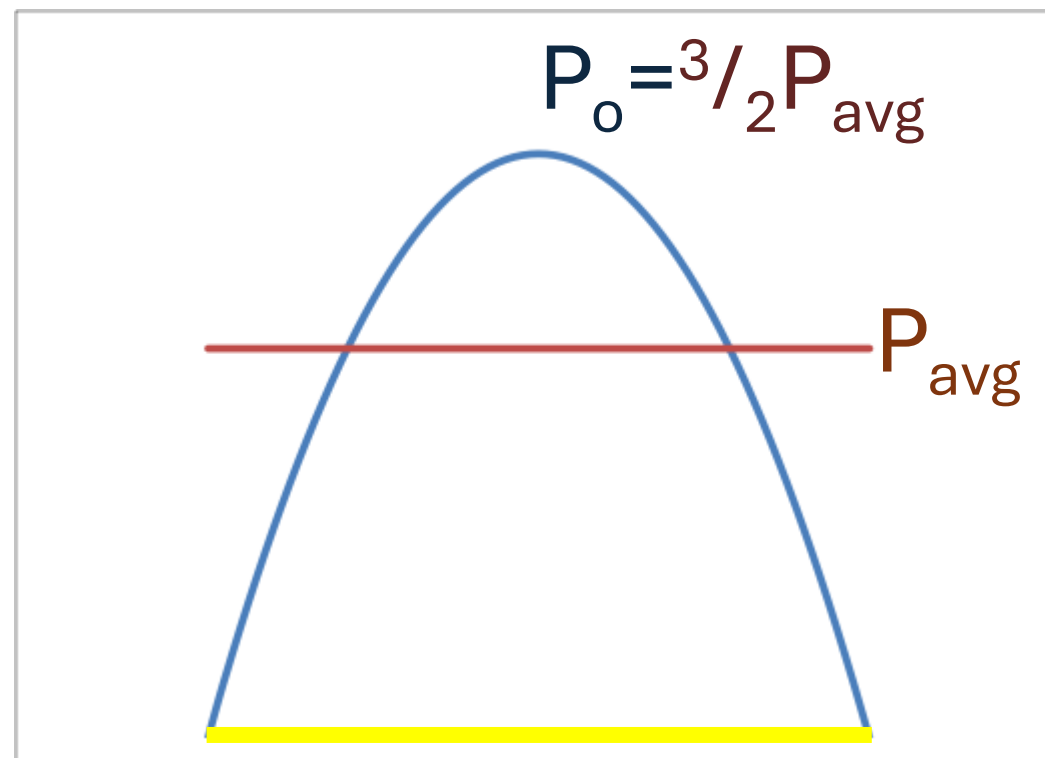
Iwnicki, S. (2006) *Handbook of Railway Dynamics*, CRC Press

Johnson, K.L. (1986) *Contact Mechanics*, Cambridge University Press



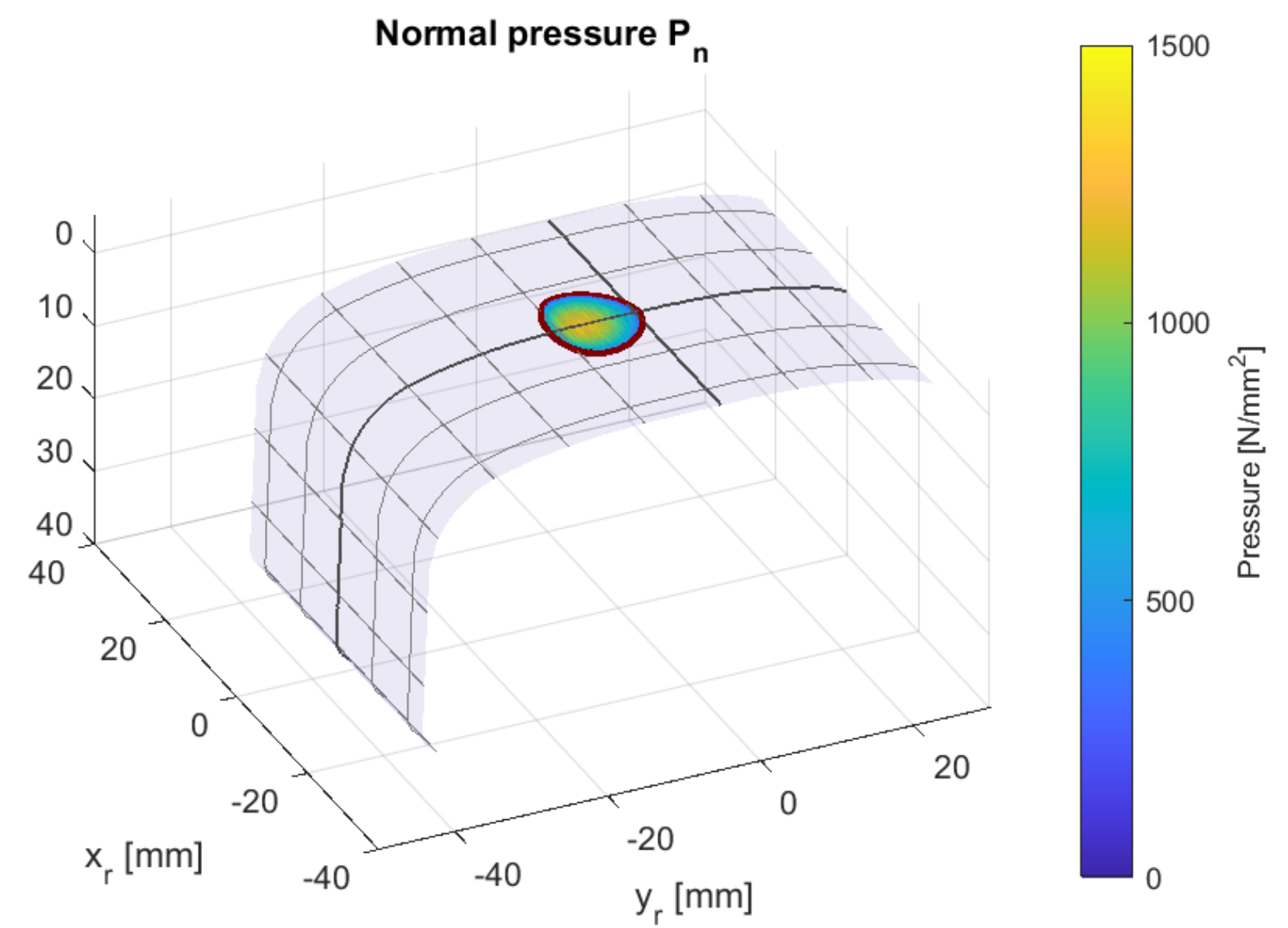
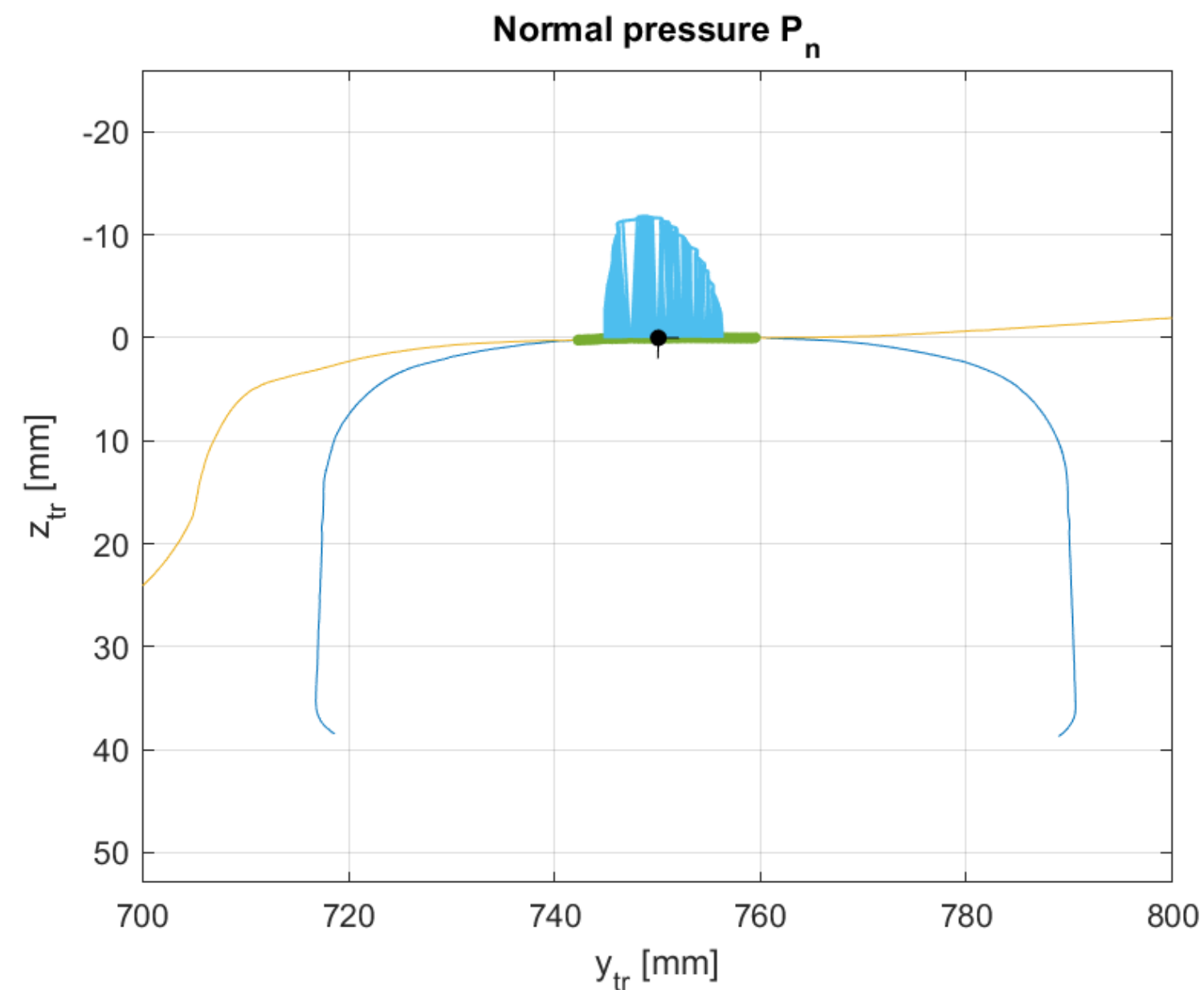
Hertzian Contact

- Hertzian Contact (1881) describes the pressures, stresses and deformations that occur when curved elastic bodies are brought into contact.
- “Contact Patches” tend to be **elliptical**
- This yields **parabolic** contact pressures



- Contact theory was subsequently broadened to apply to rolling contact (Carter and Fromm) with non-elliptical contact and arbitrary creepage (Kalker)

(Quasi) Hertzian Contact in Practice

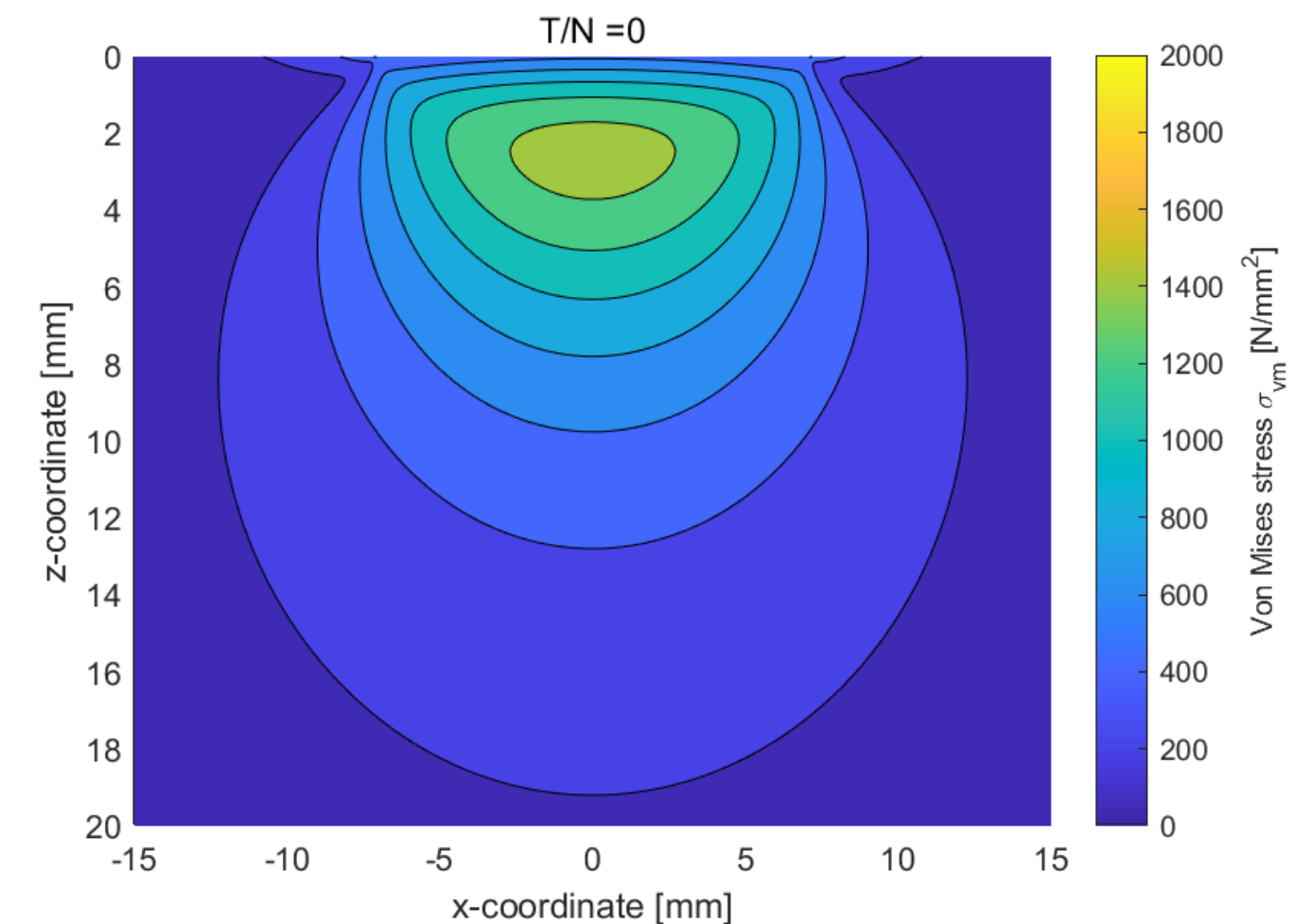
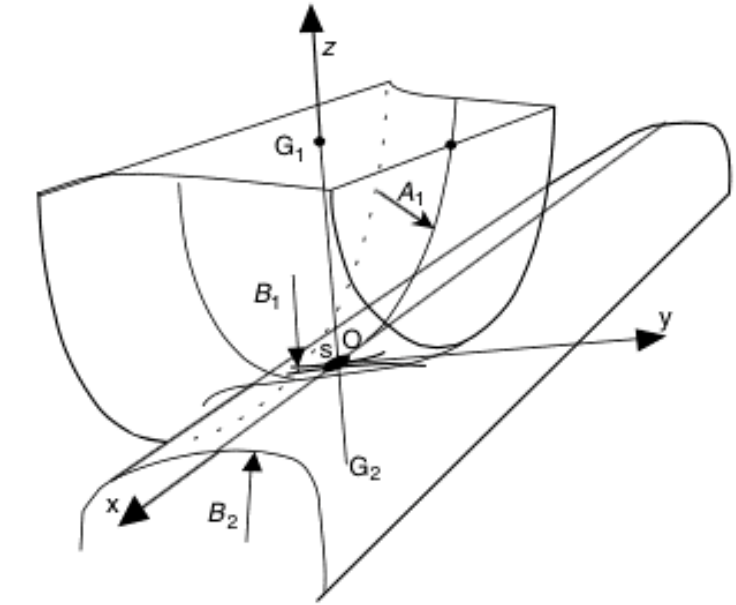


Vollebregt, E A H. "User Guide for CONTACT, Rolling and Sliding Contact with Friction." Technical Report, 2021.



Hertzian Contact and Subsurface Stresses

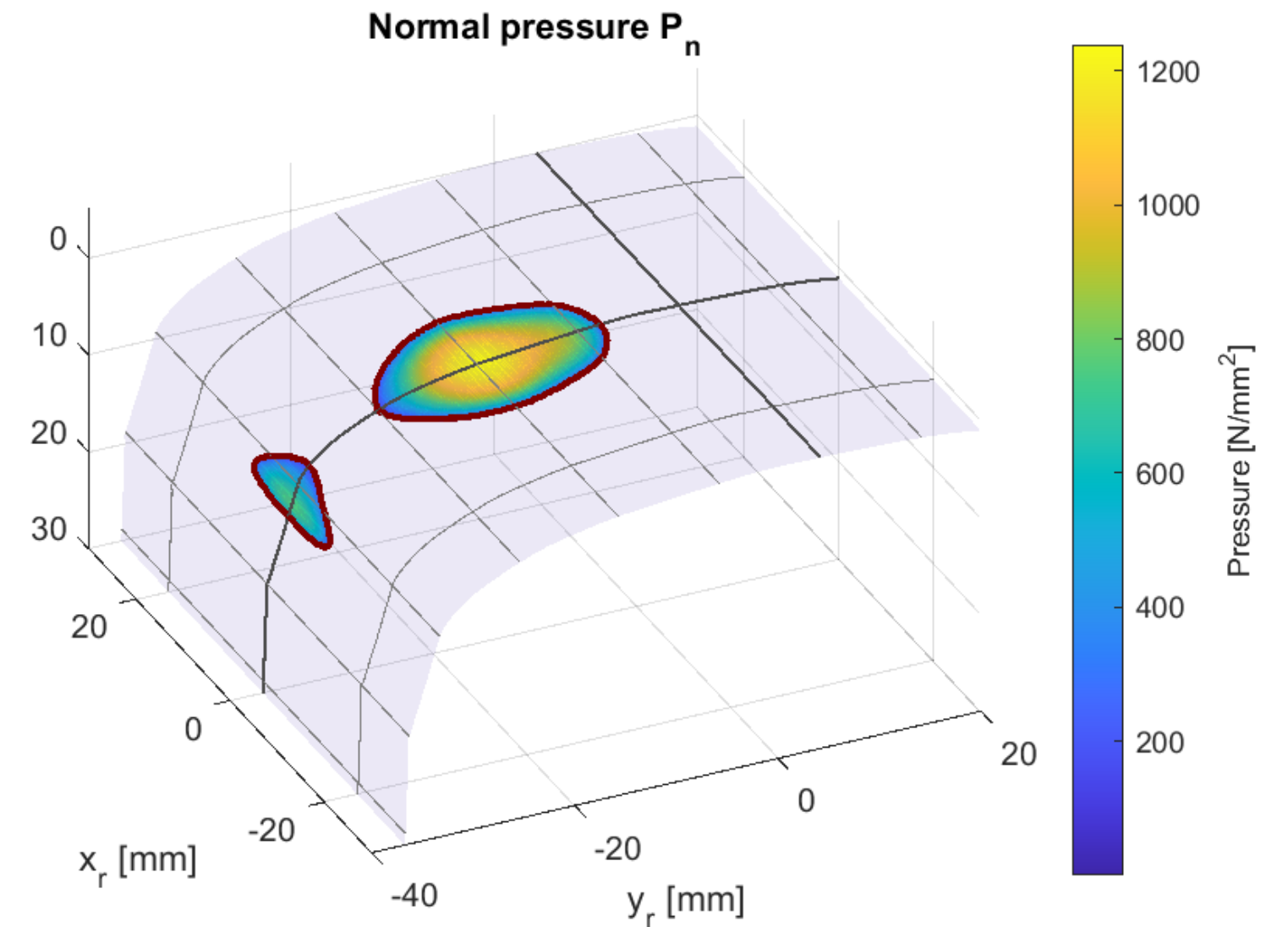
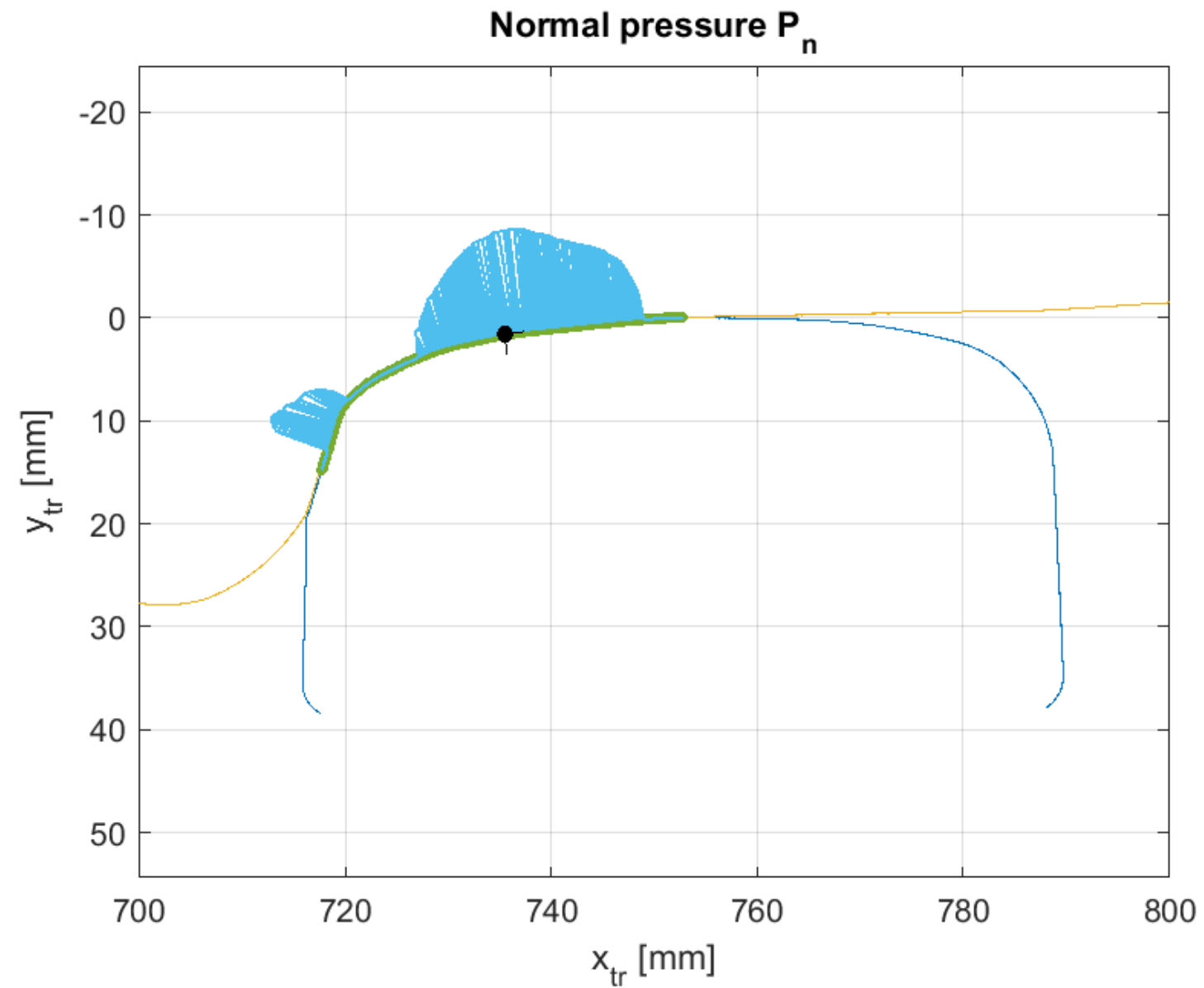
- Example, using the *CONTACT Library* to calculate the characteristic (Von Mises) stresses for the following steel-on-steel contact conditions:
 - Radius of curvature of body 1: 0.5m (20”), aka the “wheel”
 - Radius of curvature of body 2: 0.25m (10”), aka the “rail”
 - Applied Force: 160kN (36kips), aka the “weight”



Vollebregt, E A H. “User Guide for CONTACT, Rolling and Sliding Contact with Friction.” Technical Report, 2021.



Non-Hertzian Contact



Vollebregt, E A H. "User Guide for CONTACT, Rolling and Sliding Contact with Friction." Technical Report, 2021.





Fundamental #2: Traction & Creepage

Creepage at the Wheel/Rail Interface

- Creepage within the contact patch can include three types:
 - Lateral
 - Longitudinal
 - Spin



What does Longitudinal Creepage *mean*?...



Coasting along a flat and dry highway

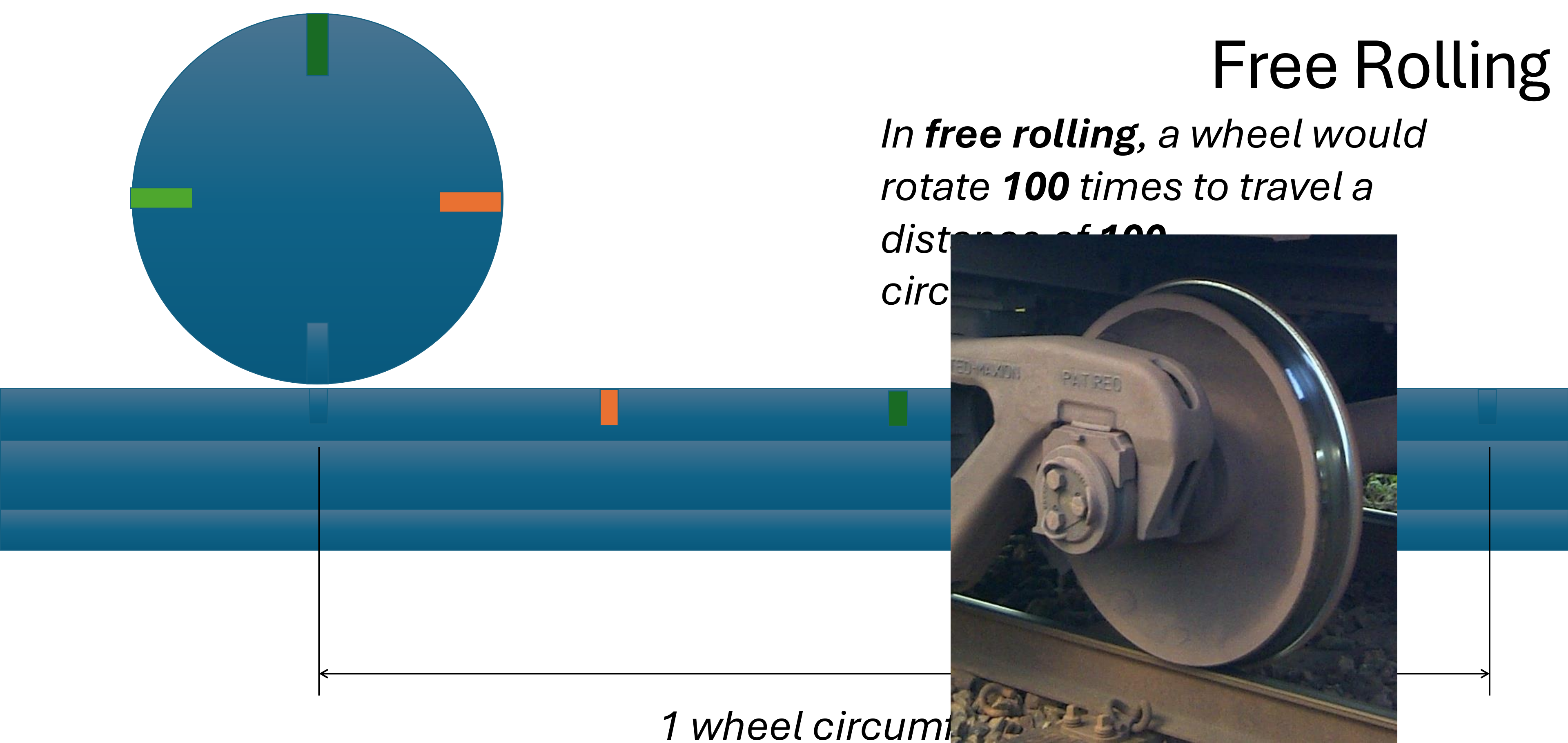


Climbing a steep and snowy mountain road



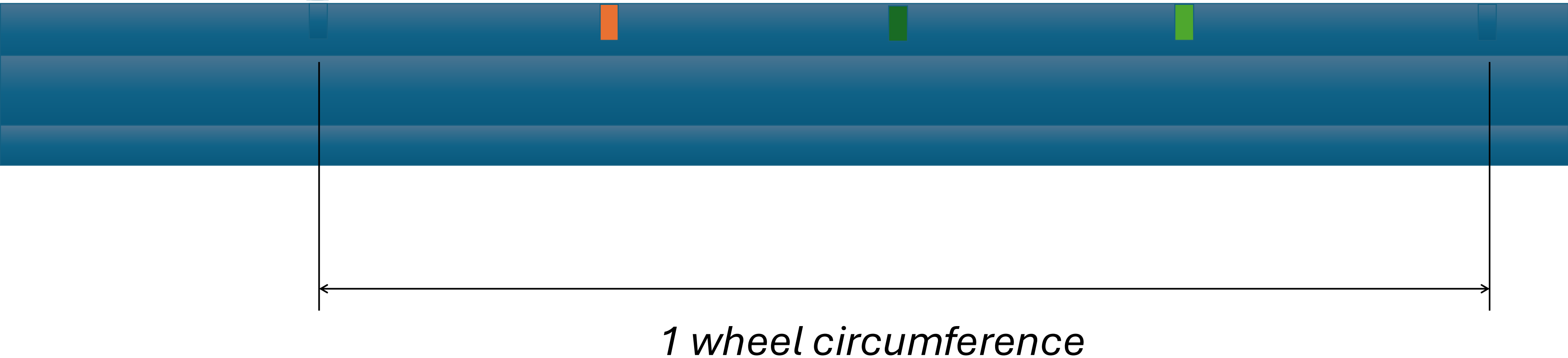
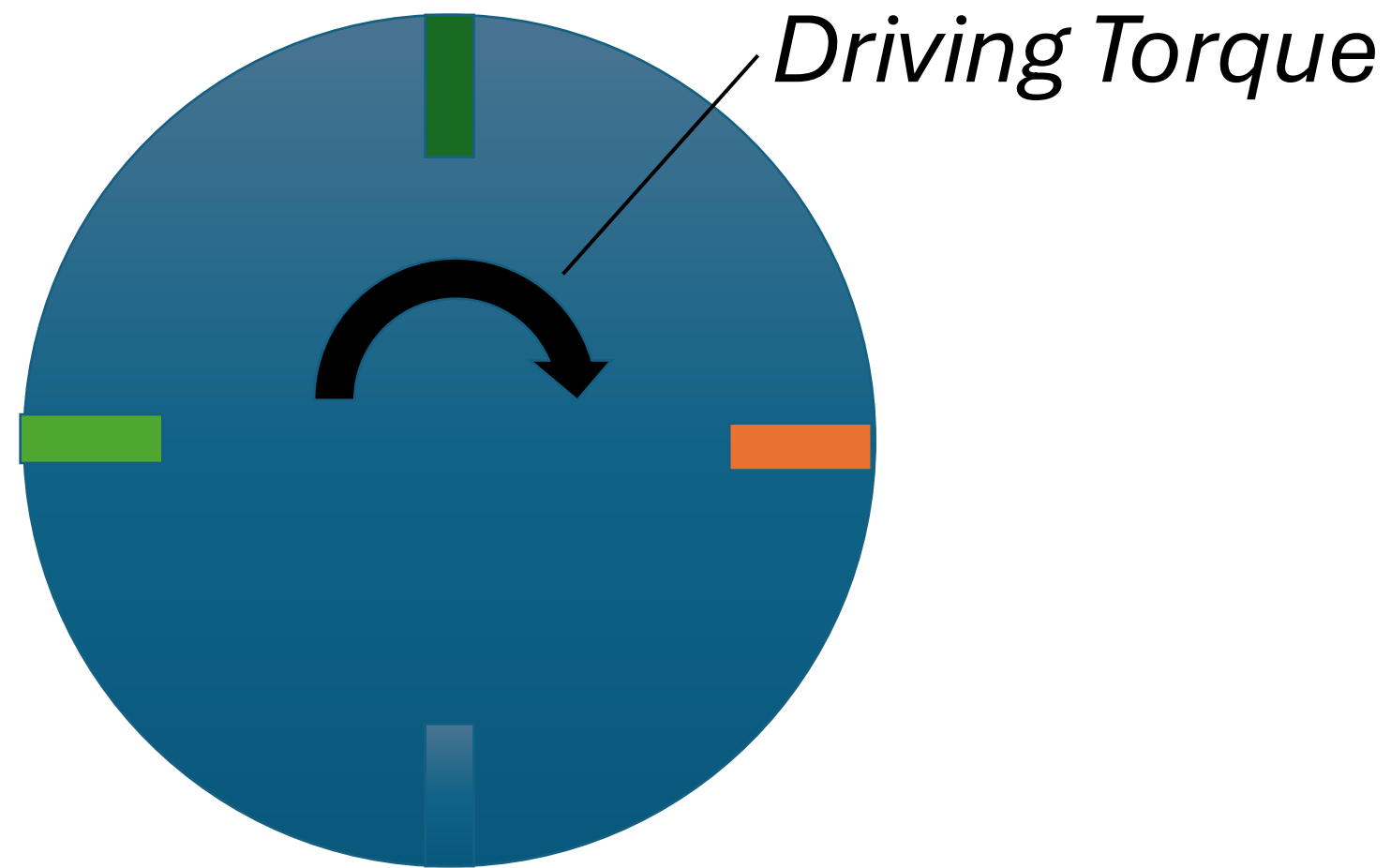
Free Rolling

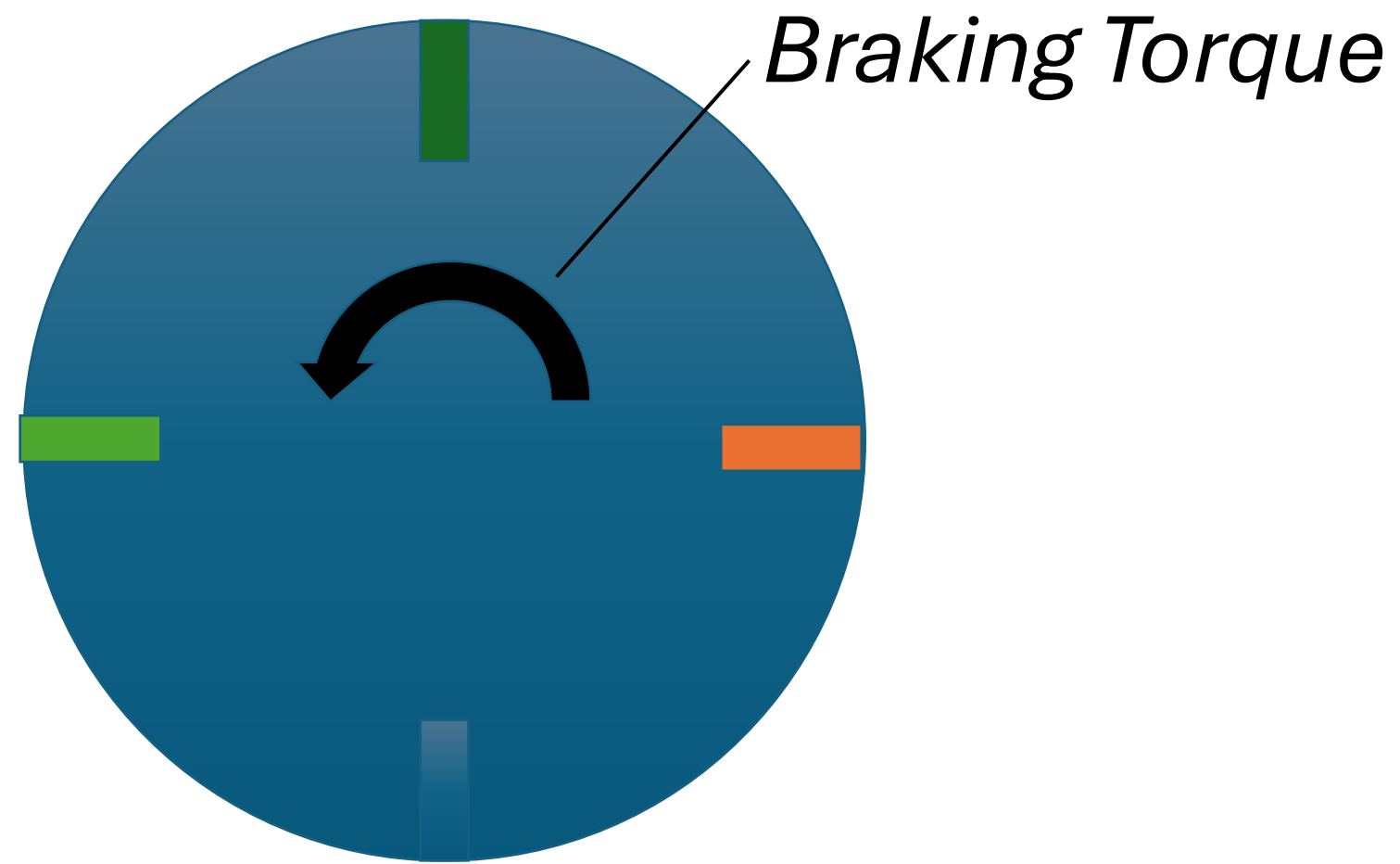
*In **free rolling**, a wheel would rotate **100** times to travel a distance of **100** circles*



Positive (Longitudinal) Creepage

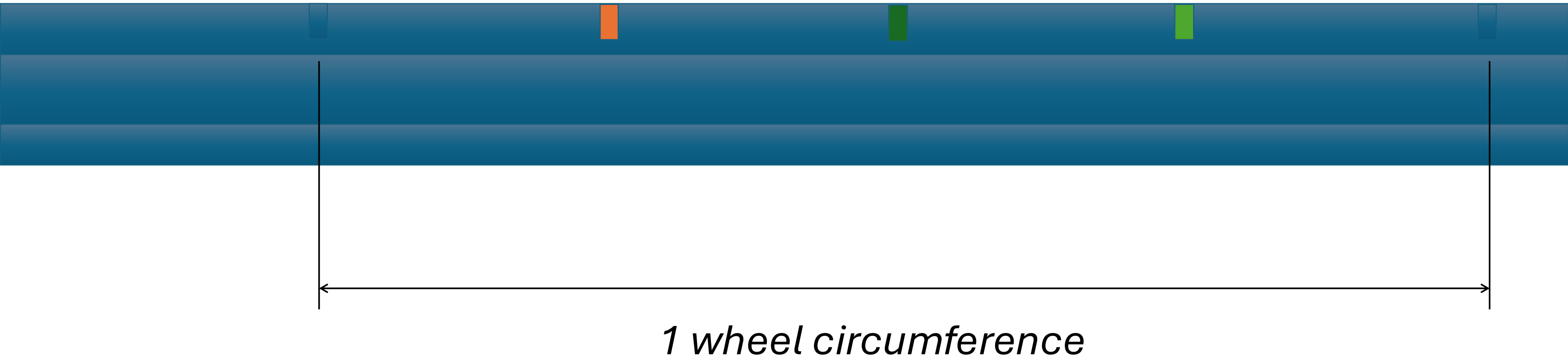
*At 1% **positive** creepage, a wheel would rotate **101** times to travel a distance of **100** circumferences.*



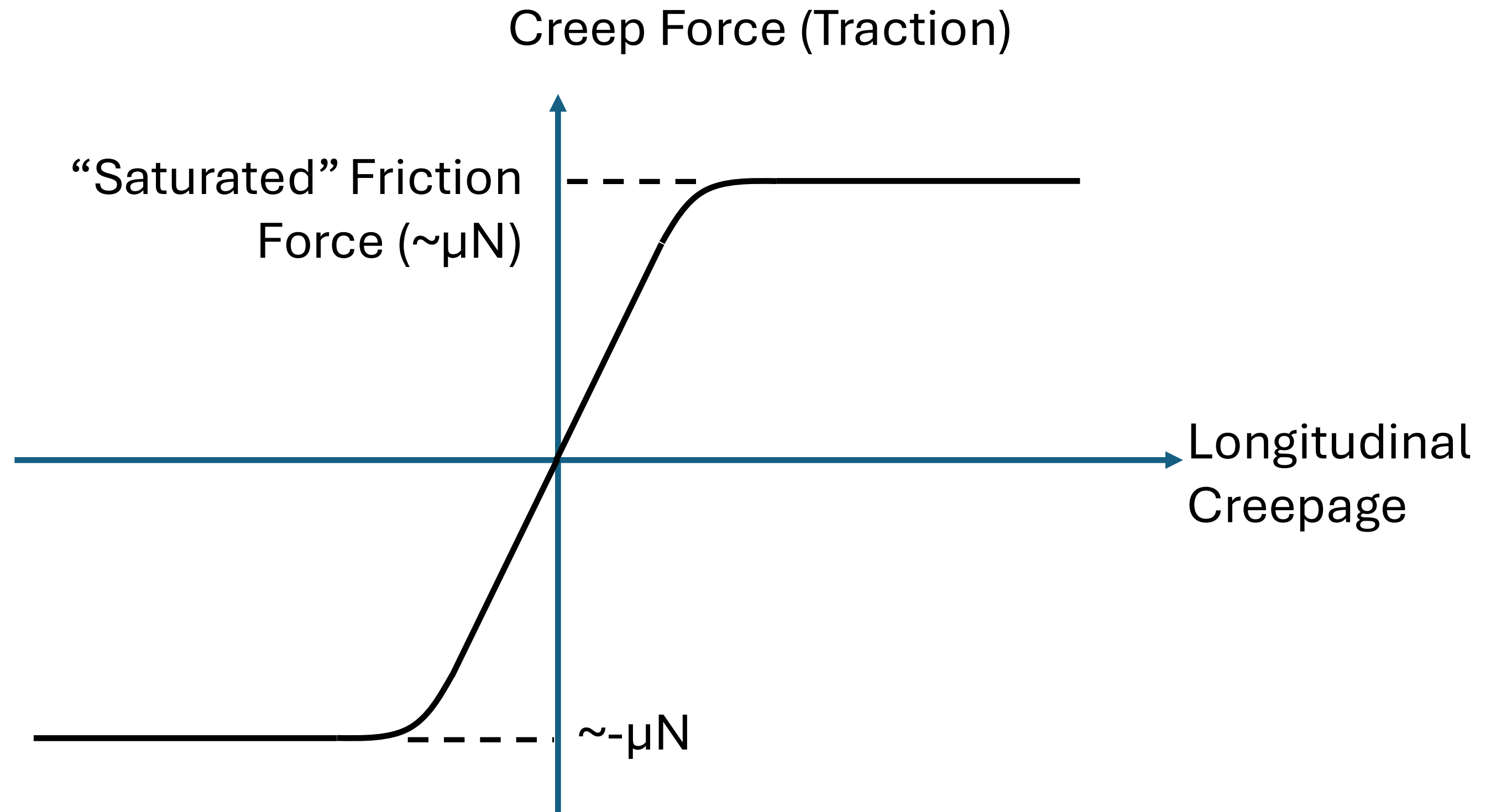


Negative (Longitudinal) Creepage

*At 1% **negative** creepage, a wheel would rotate **99** times to travel a distance of **100** circumferences.*

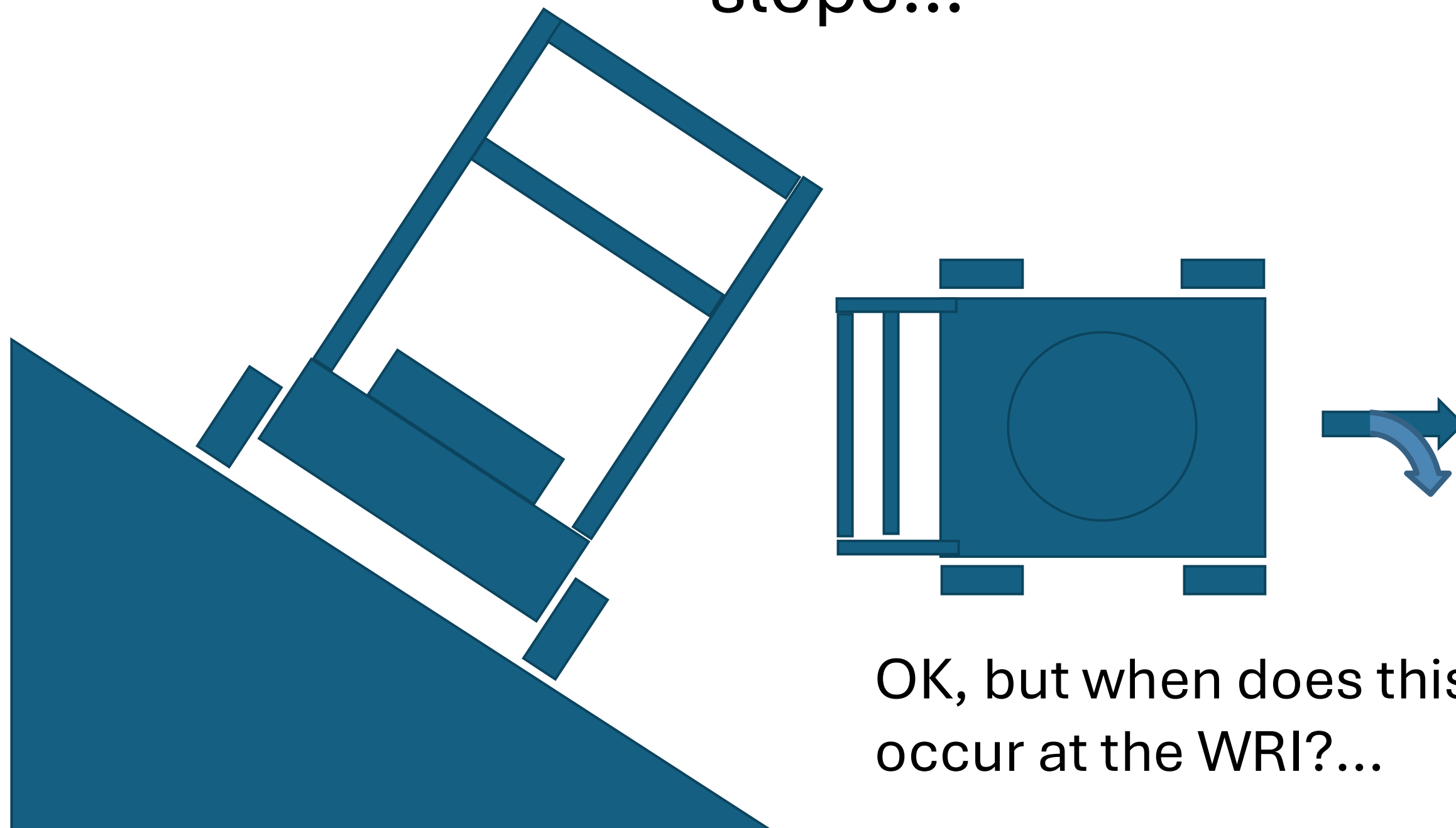


The Traction-Creepage Curve



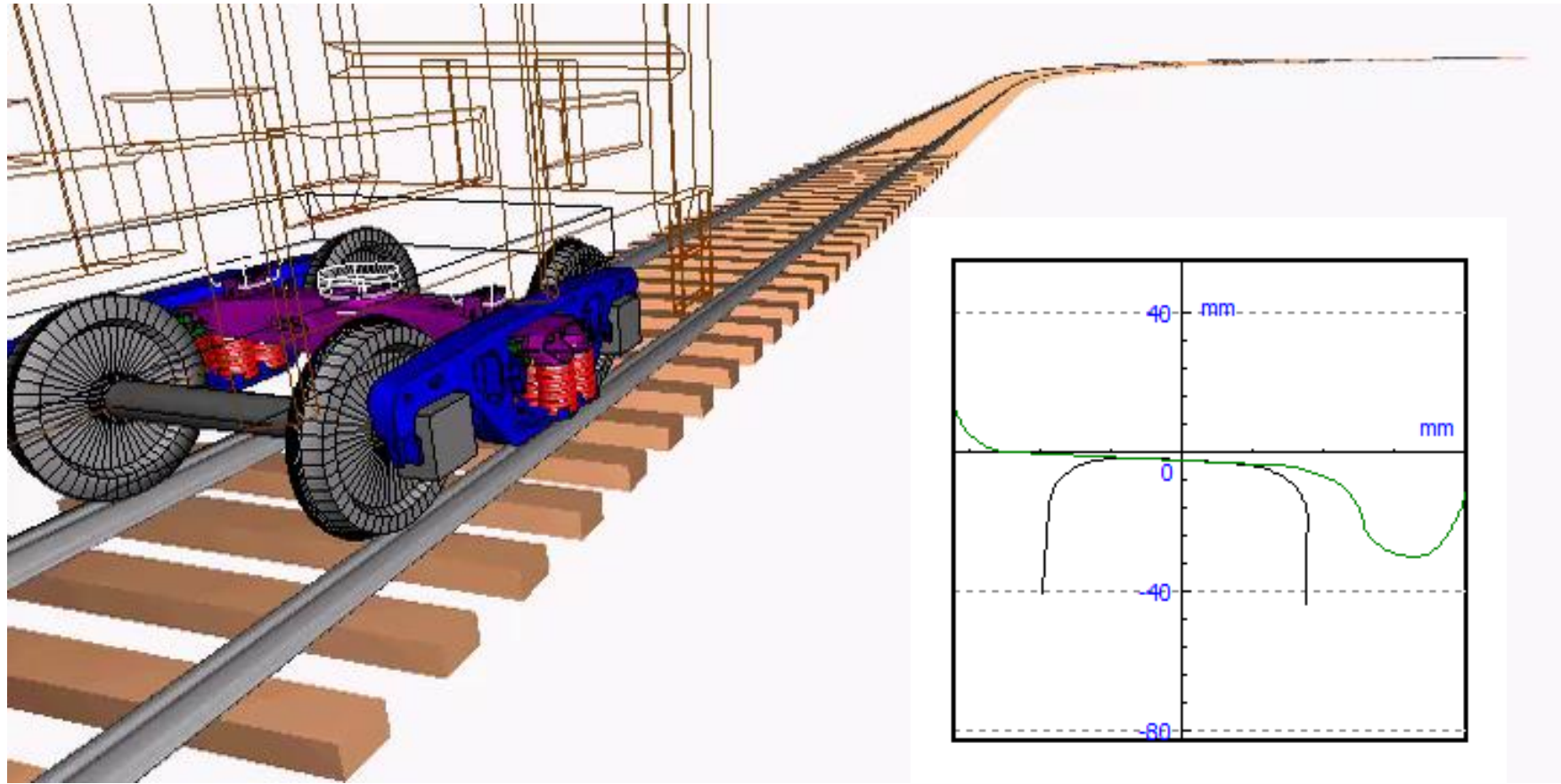
Lateral creepage

Imagine pushing a lawnmower across a steep slope...



OK, but when does this occur at the WRI?...

Vehicle Curving (UM Simulation)

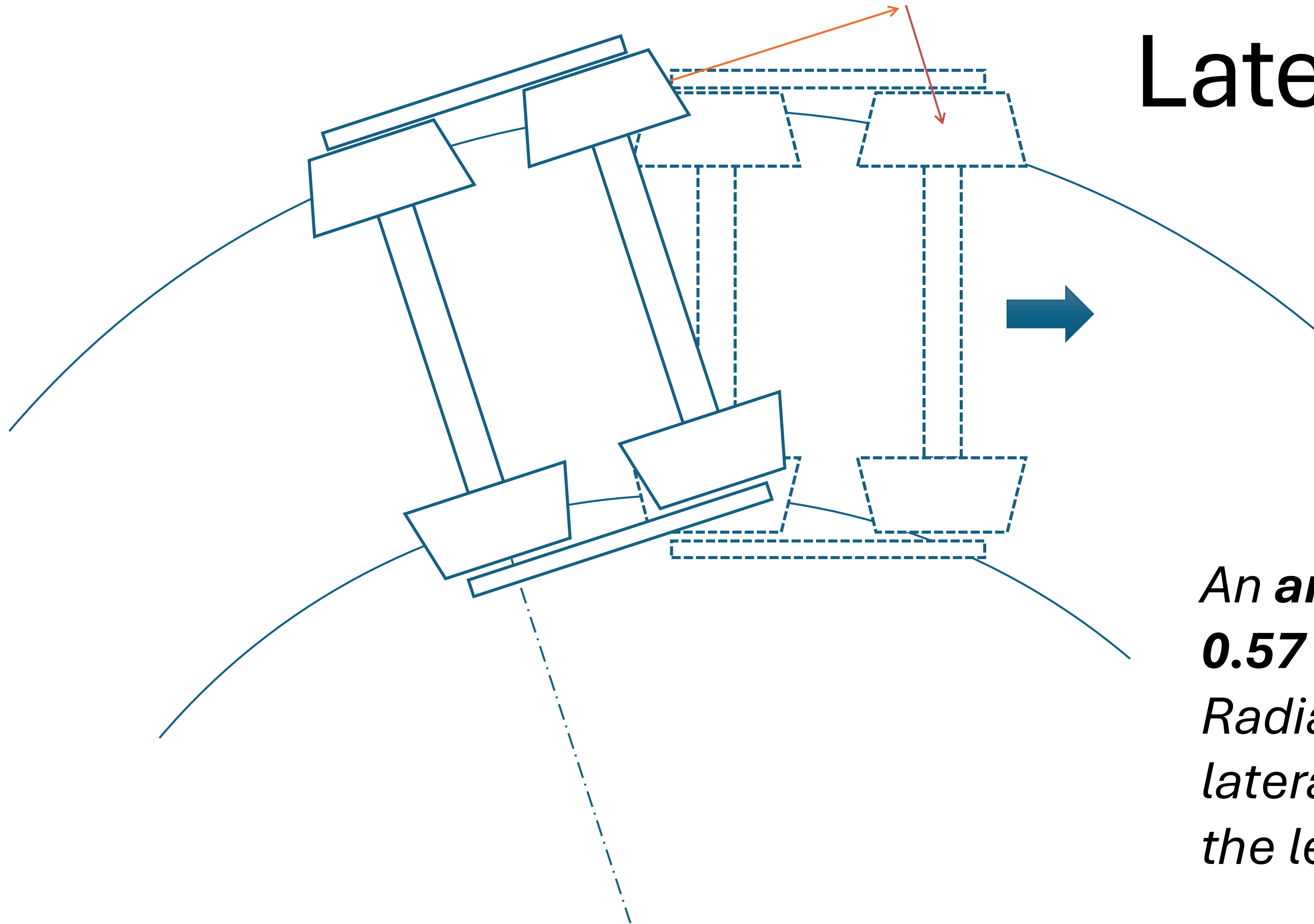


Leading Wheelset

Trailing Wheelset



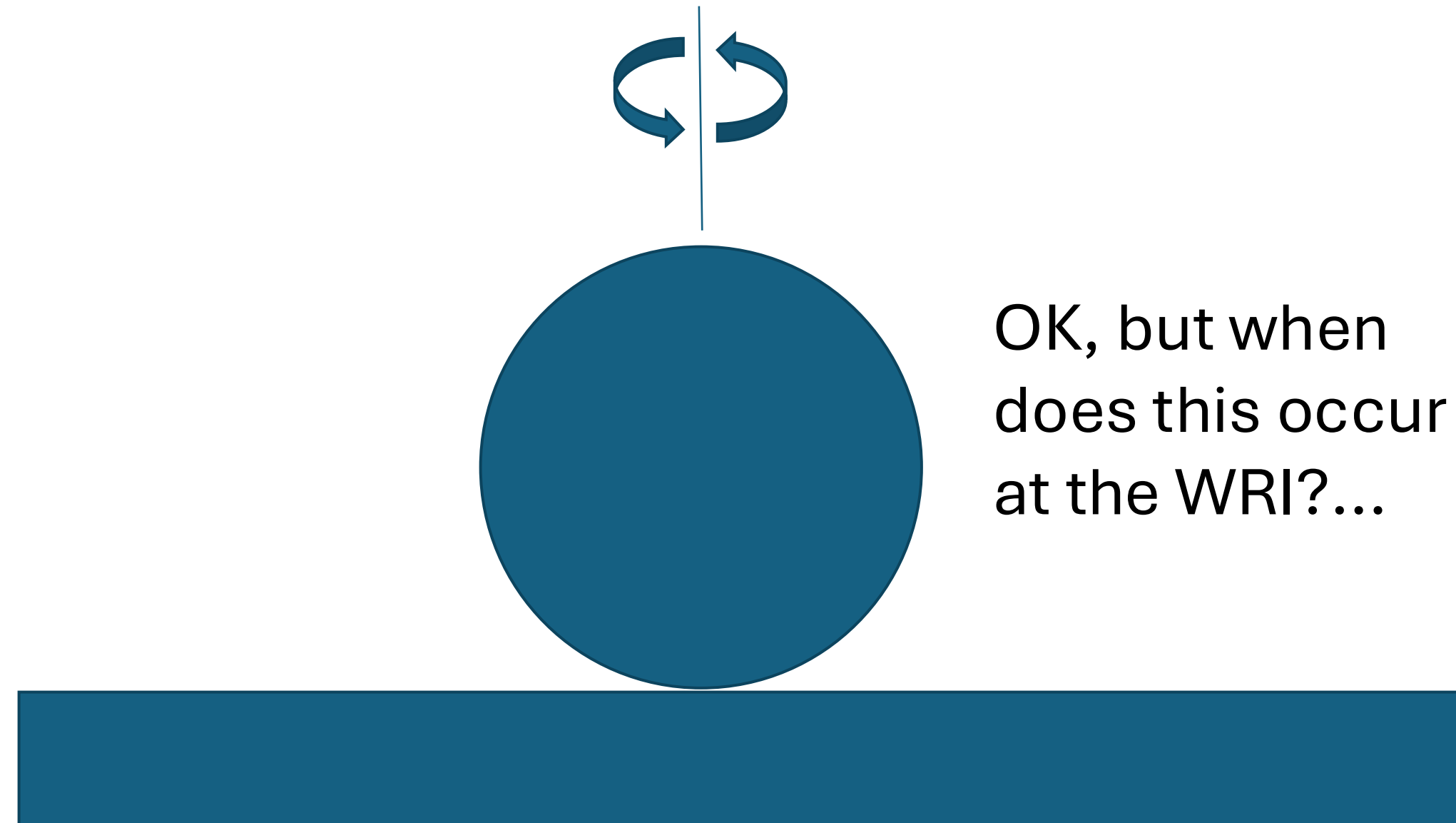
Lateral Creepage



*An **angle of attack (AoA)** of **0.57 degrees** (0.01 Radians) corresponds to a lateral creepage of **1%** at the leading wheelset.*

Spin Creepage

Think of spinning a coin on a tabletop....

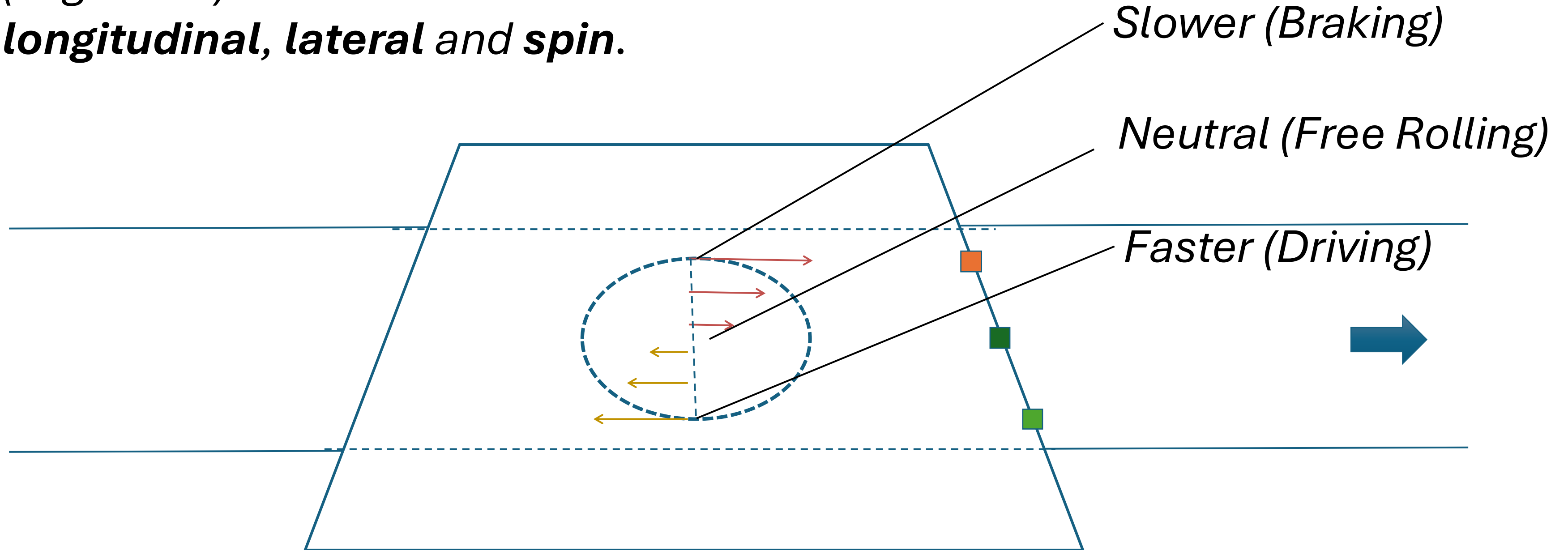


OK, but when does this occur at the WRI?...

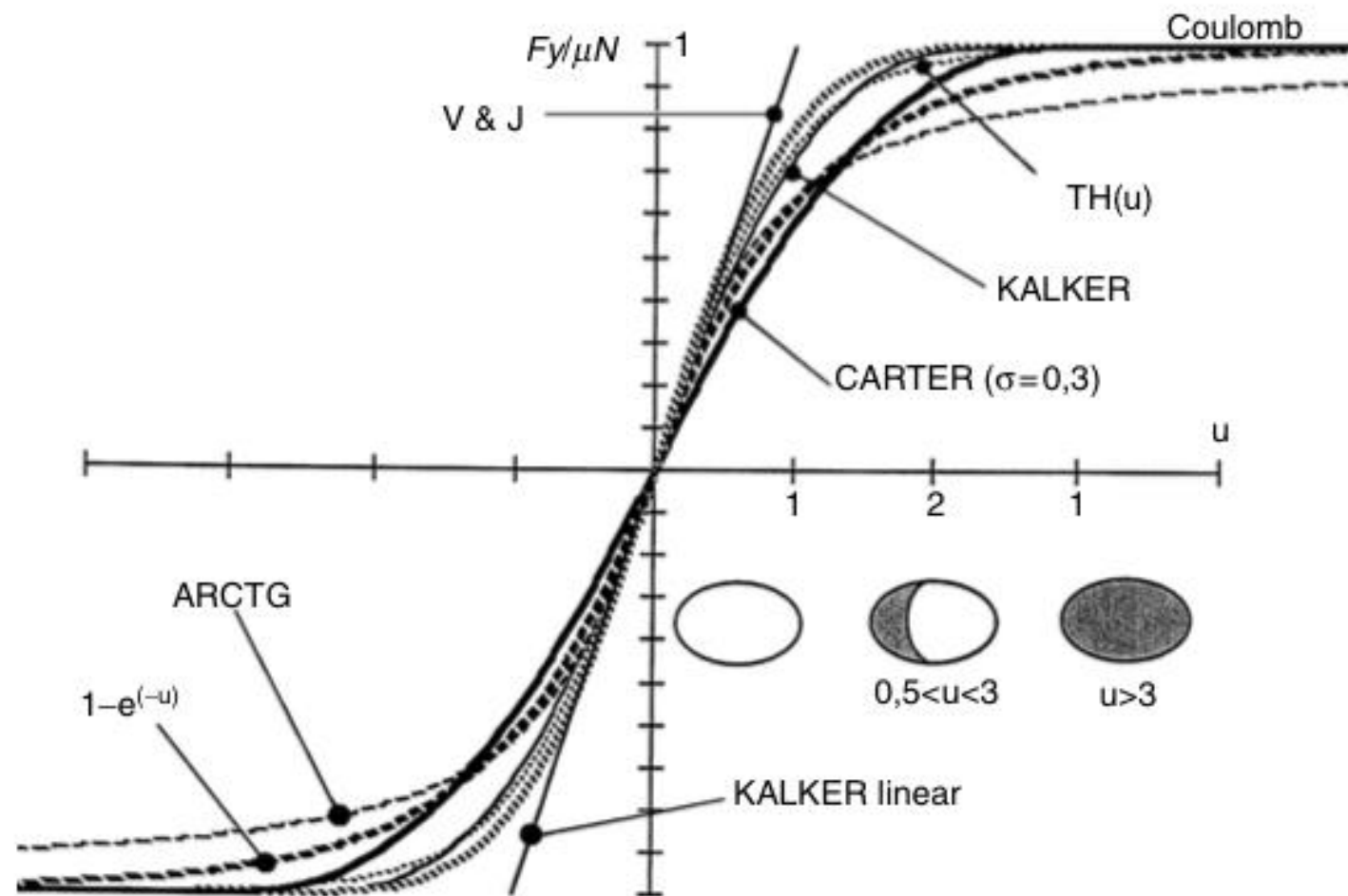


Spin Creepage

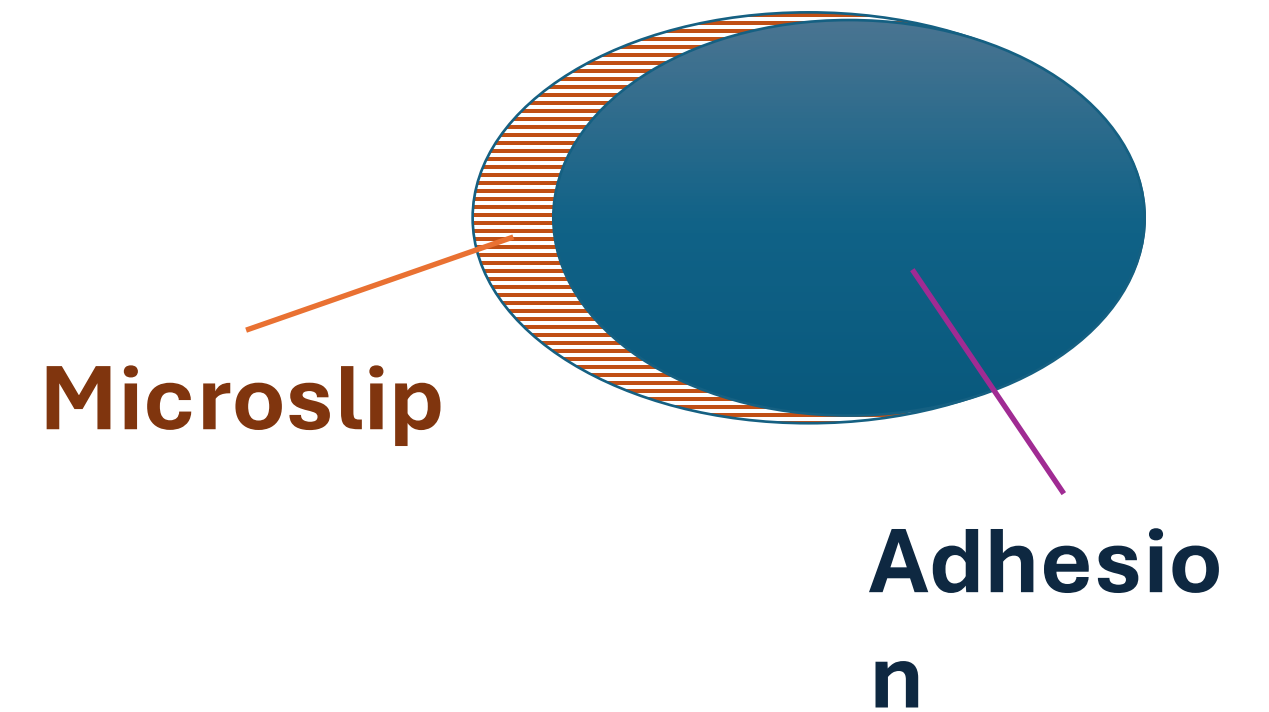
The **net creepage** vector at the wheel/rail interface is
(in general) a combination of
longitudinal, lateral and spin.



The Traction-Creepage Curve: Stick and Slip



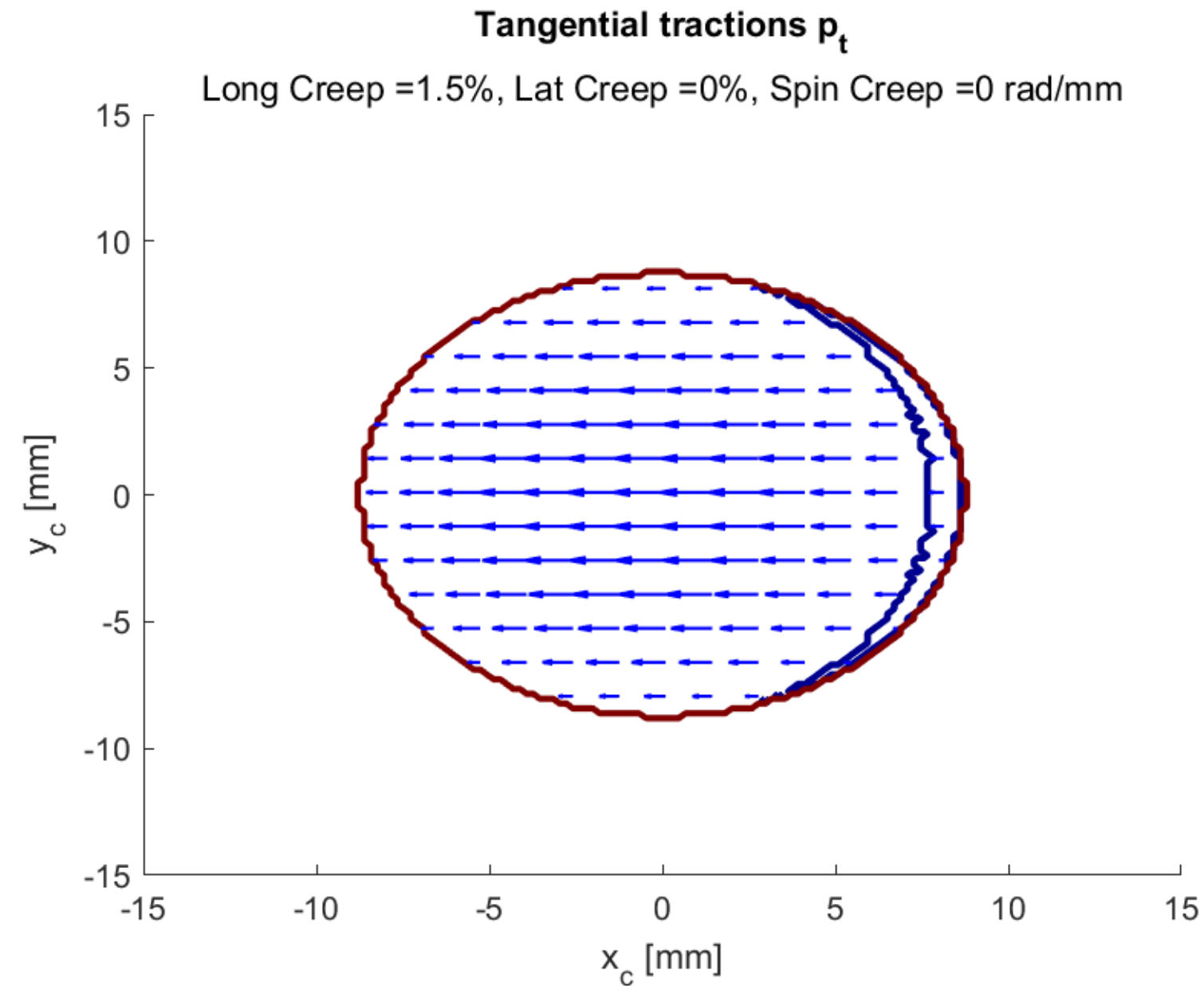
“Heuristic” expressions used for the saturation and physical meaning of the different parts.



Iwnicki, S. (2006) Handbook of Railway Dynamics, CRC Press

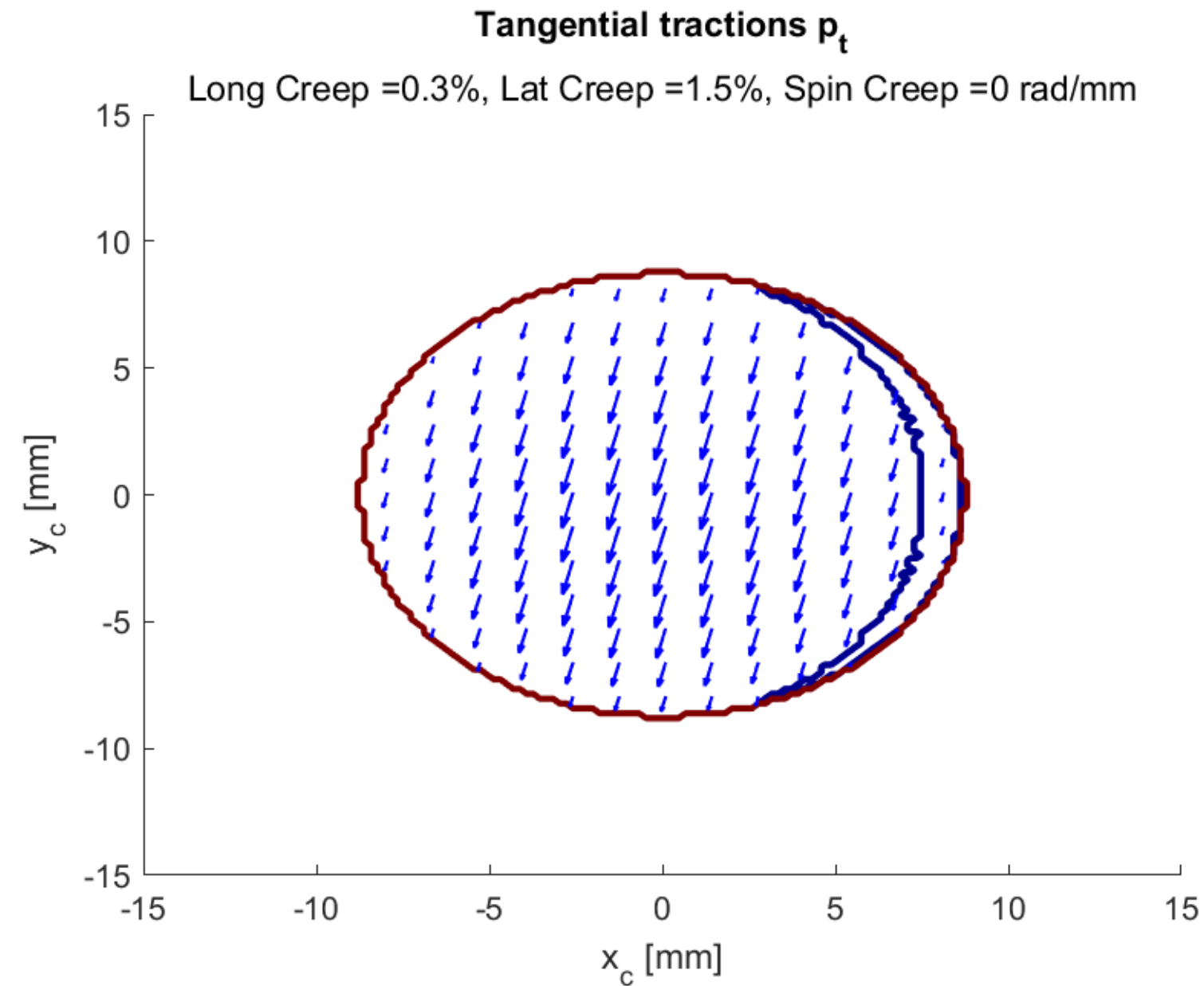


Increasing Longitudinal Creepage in the Contact Patch:

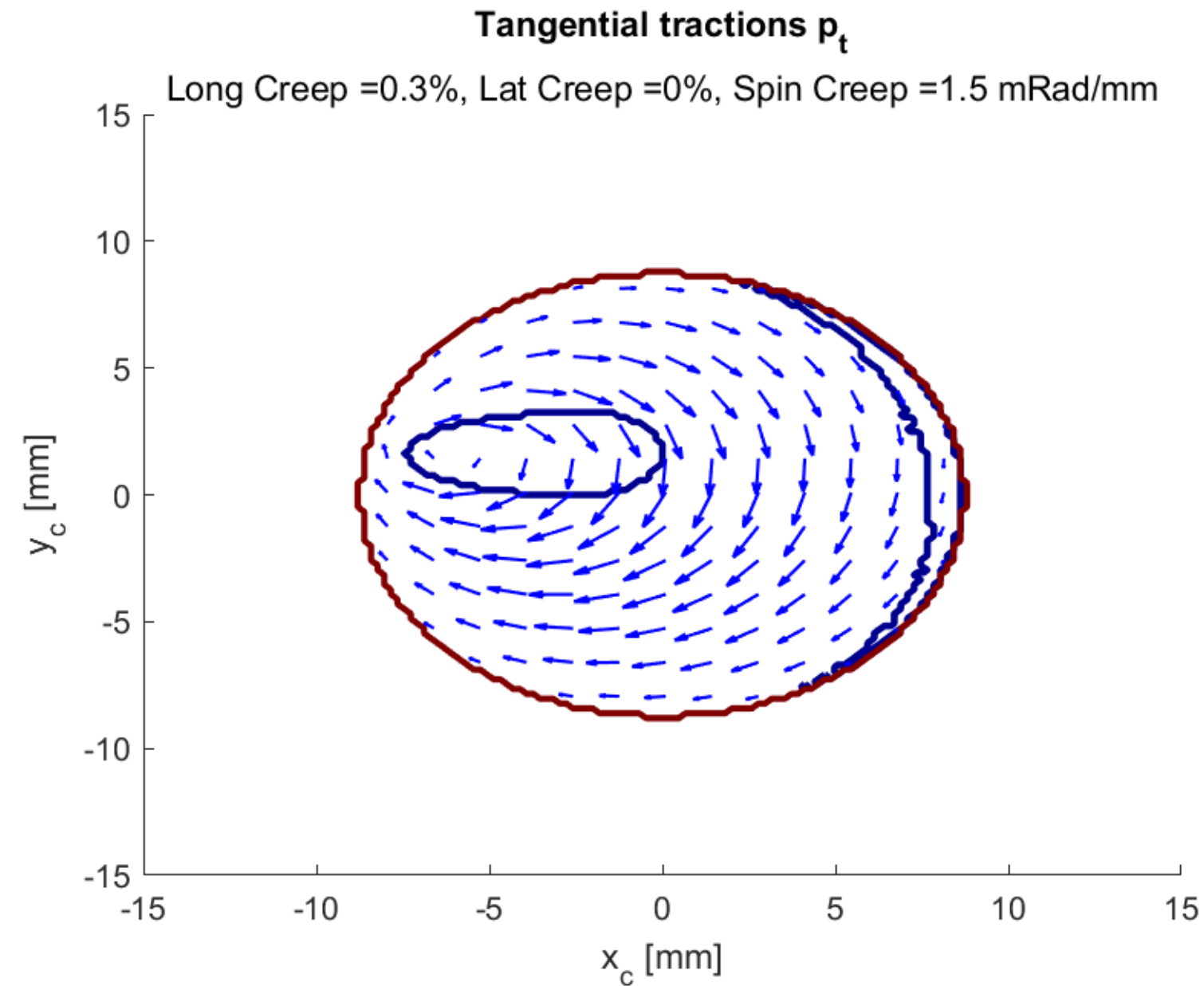


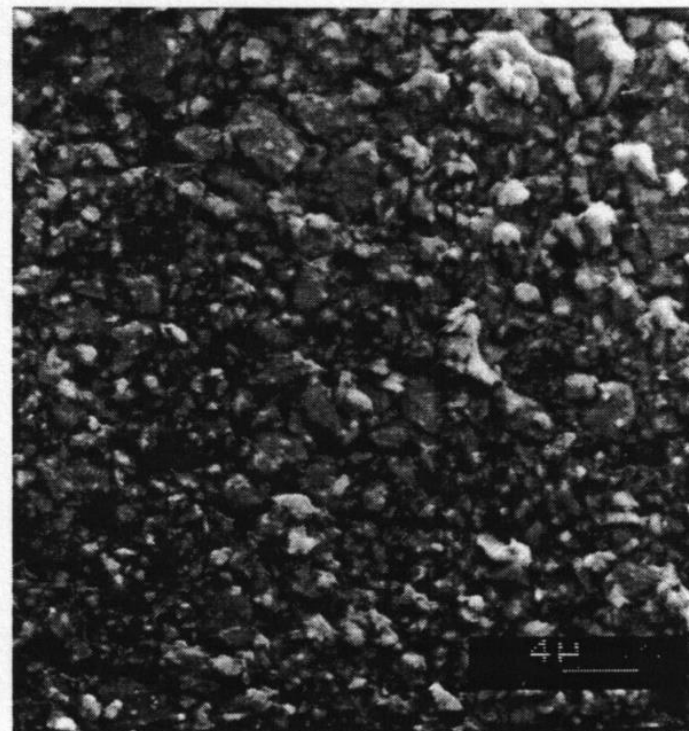
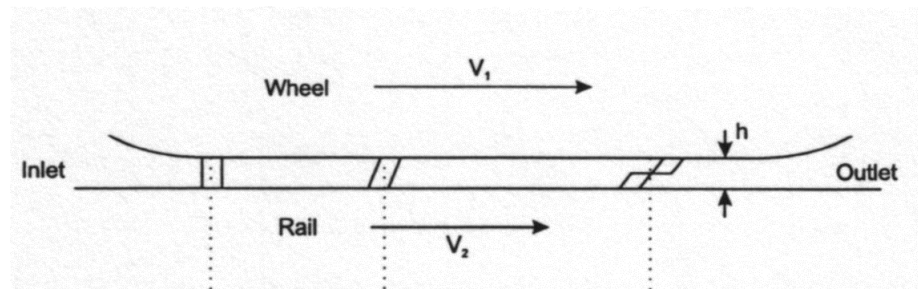


Combined Longitudinal and Lateral Creepage in the Contact Patch:



Combined Longitudinal and Spin Creepage in the Contact Patch:





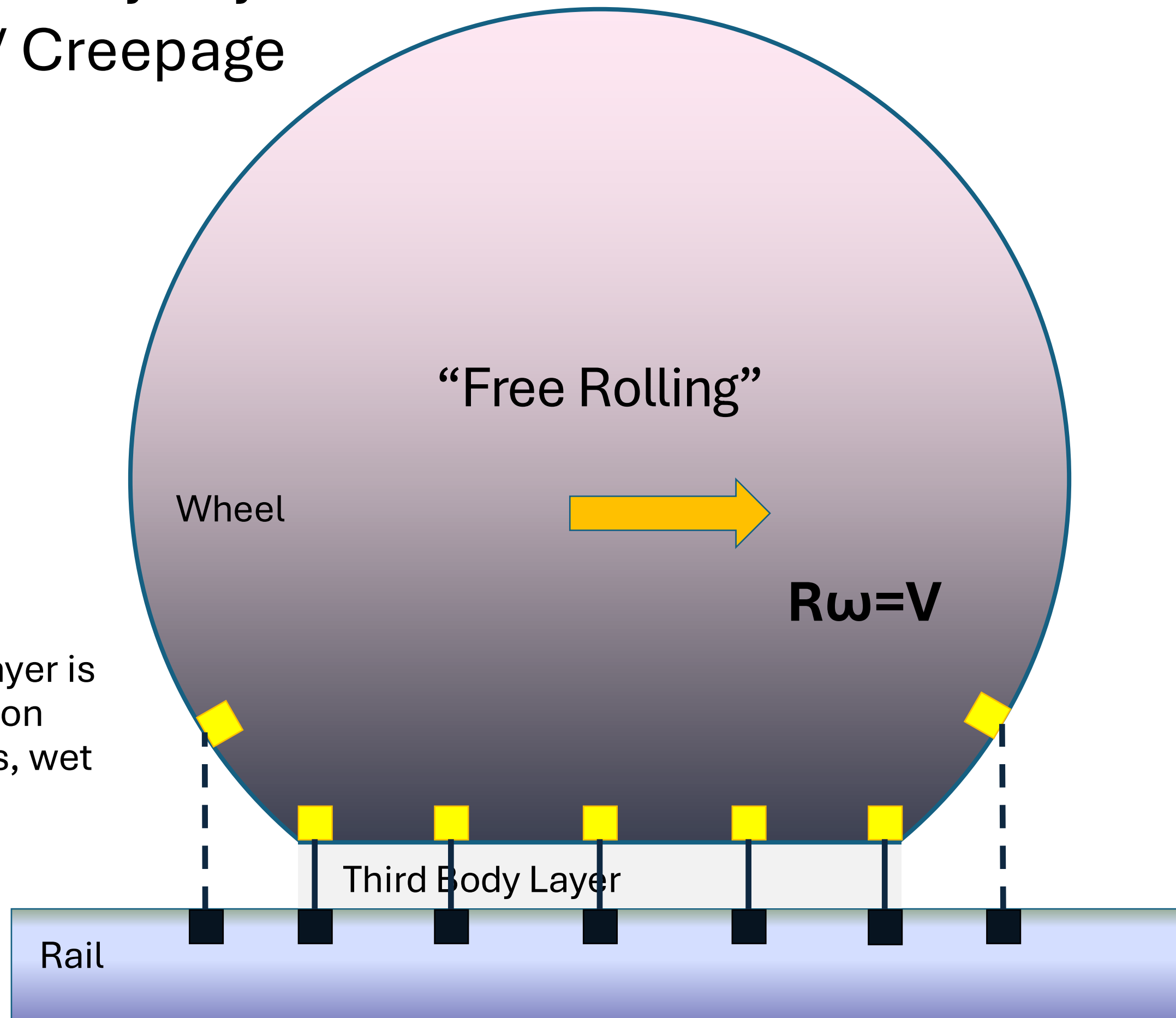
Reality Check: The Third Body Layer





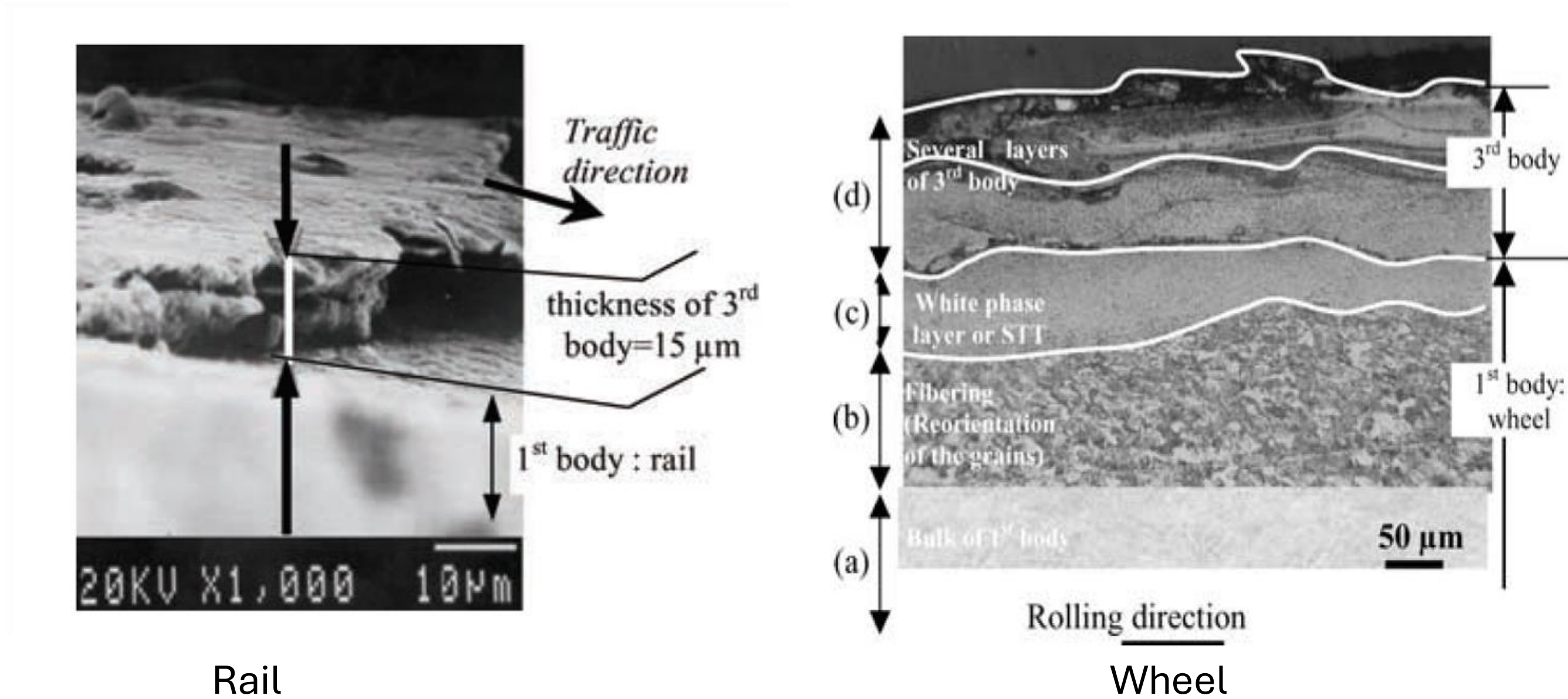
The Third Body Layer and Traction / Creepage

- Third Body Layer is made up of iron oxides, sands, wet paste, leaves etc....



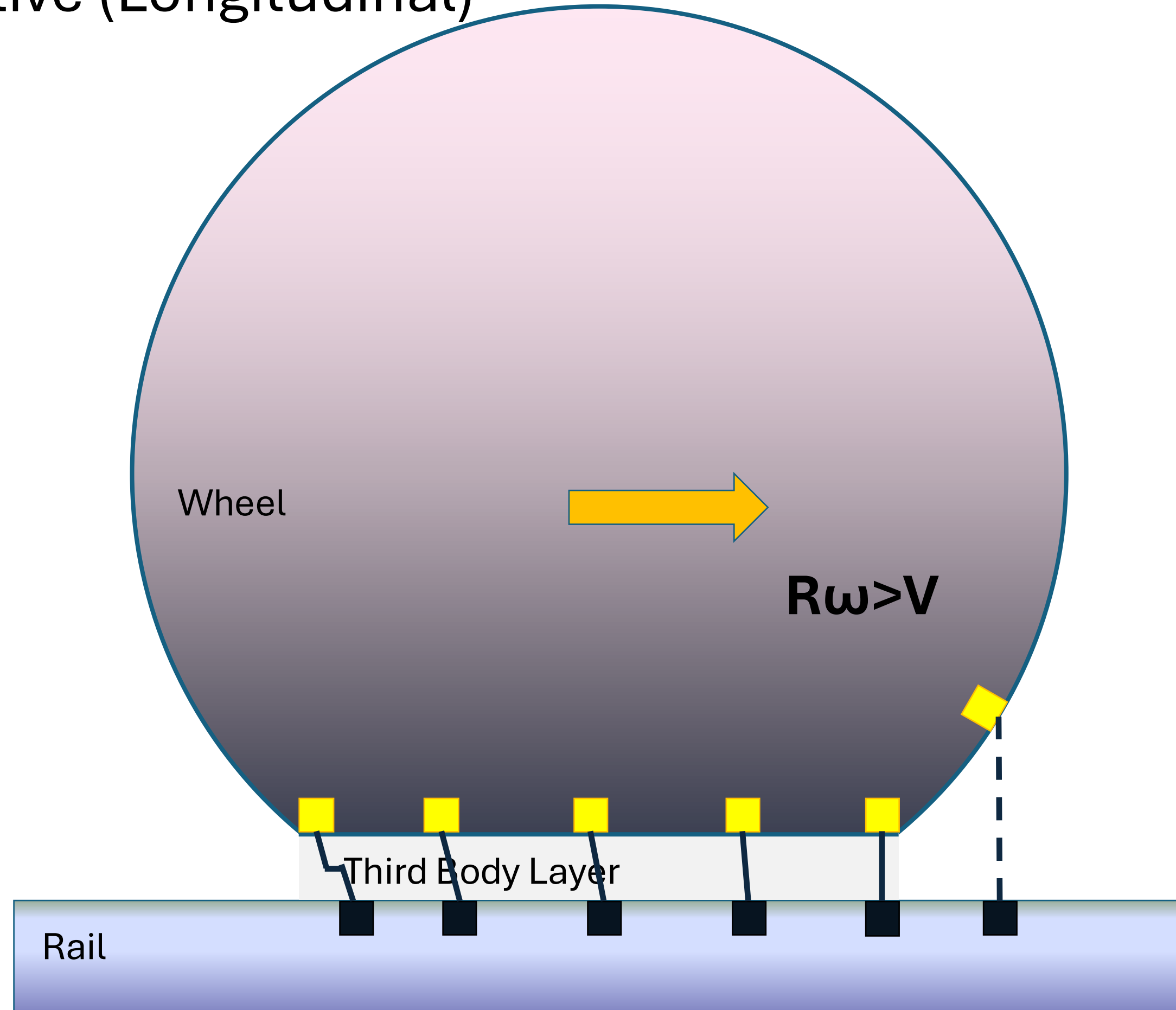


Third Body Layer – Micron Scale



Y.Berthier, S. Decartes, M.Busquet et al. (2004). The Role and Effects of the third body in the wheel rail interaction. *Fatigue Fract. Eng. Mater Struct.* 27, 423-436

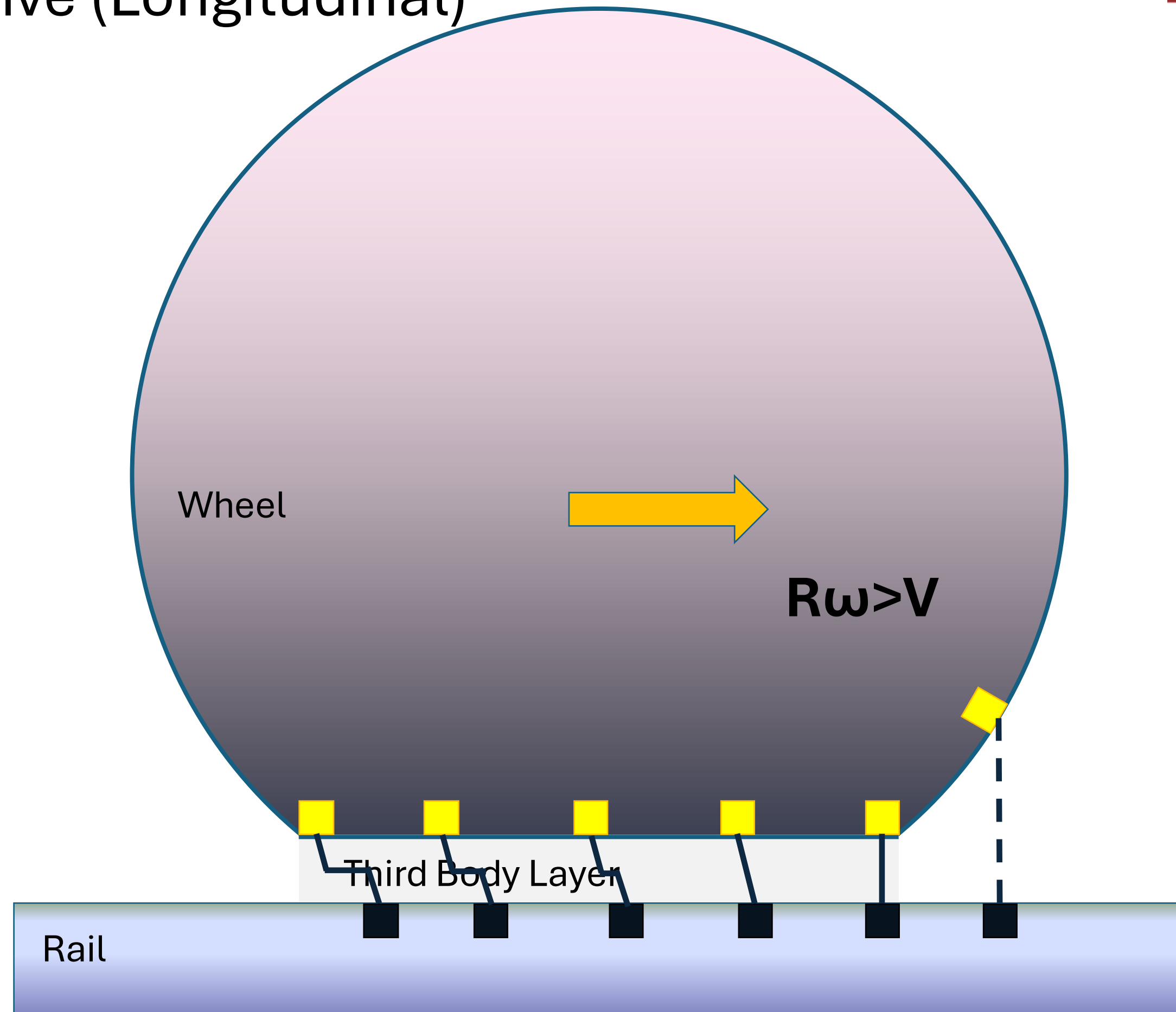
“Small” Positive (Longitudinal) Creepage



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“Large” Positive (Longitudinal) Creepage

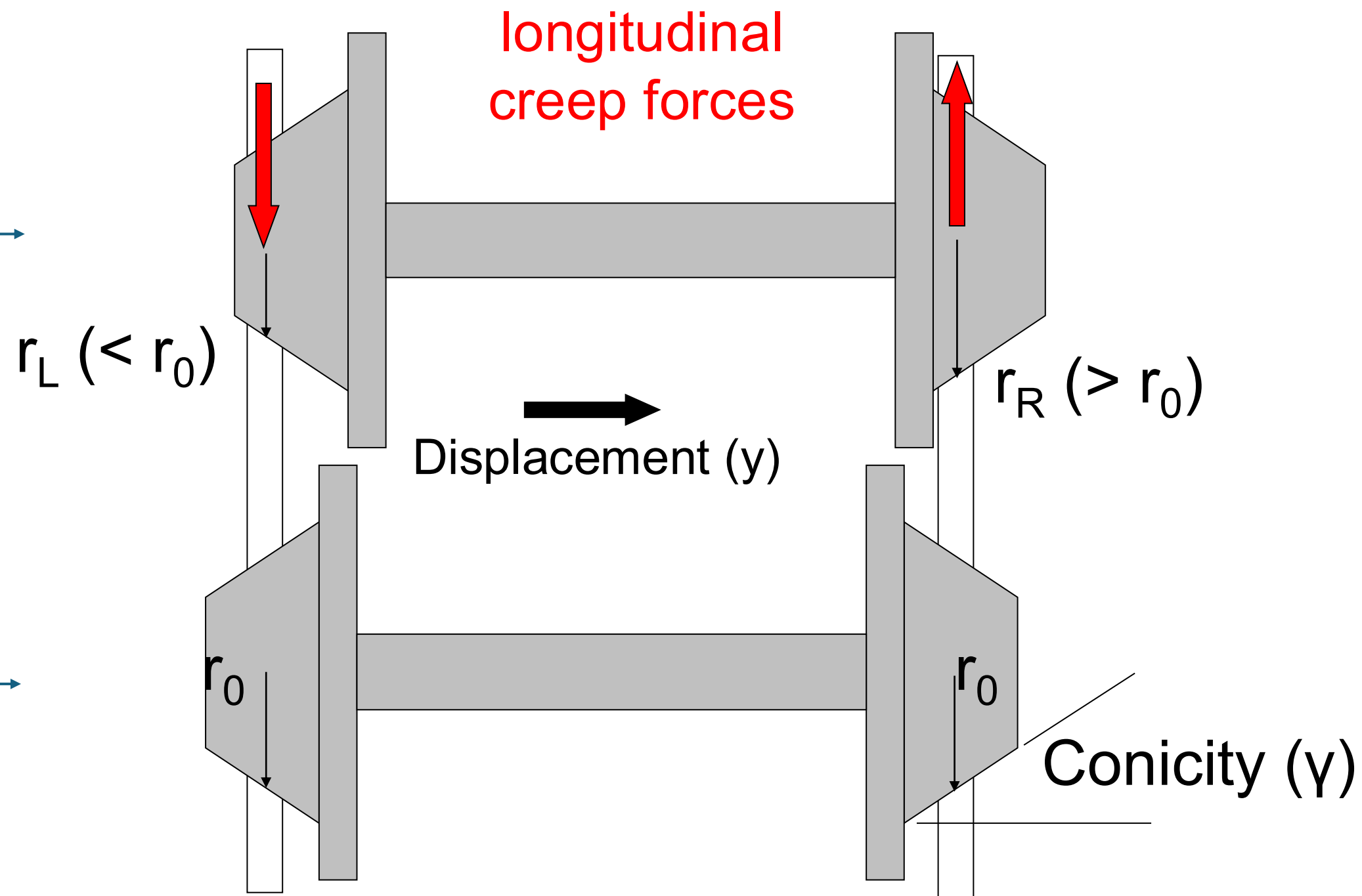
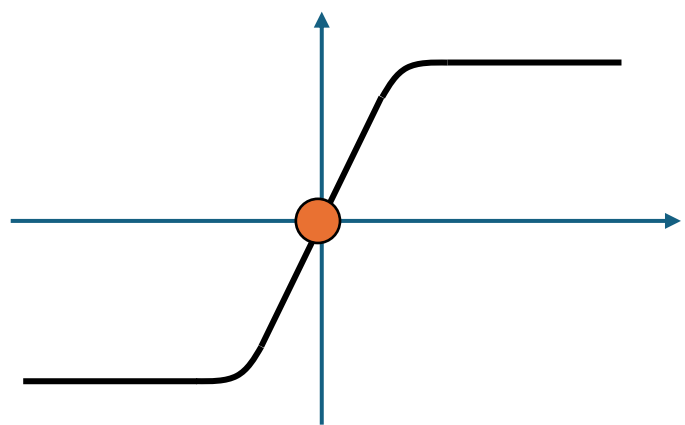
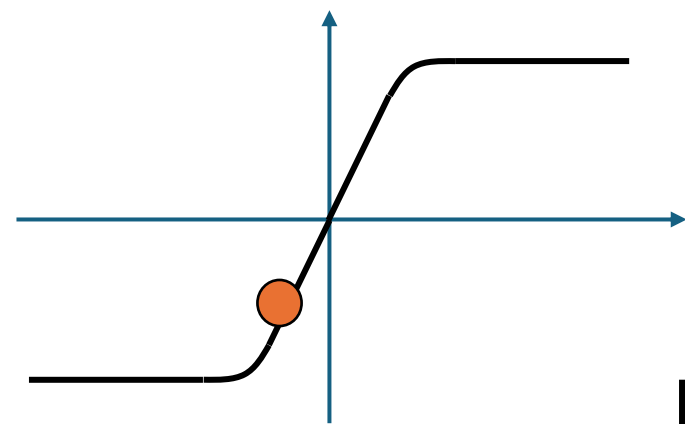




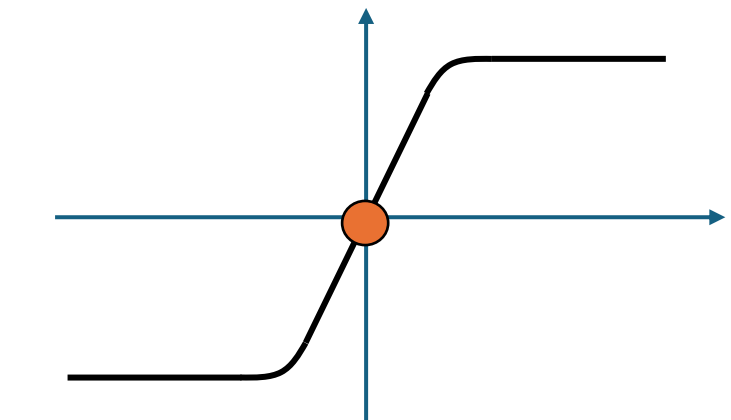
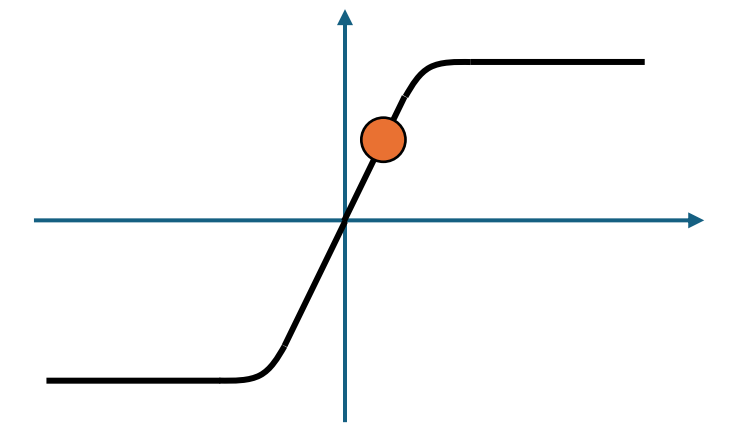
Steering Forces: Putting it all together

The Wheelset and Steering Forces

longitudinal
traction/creepage



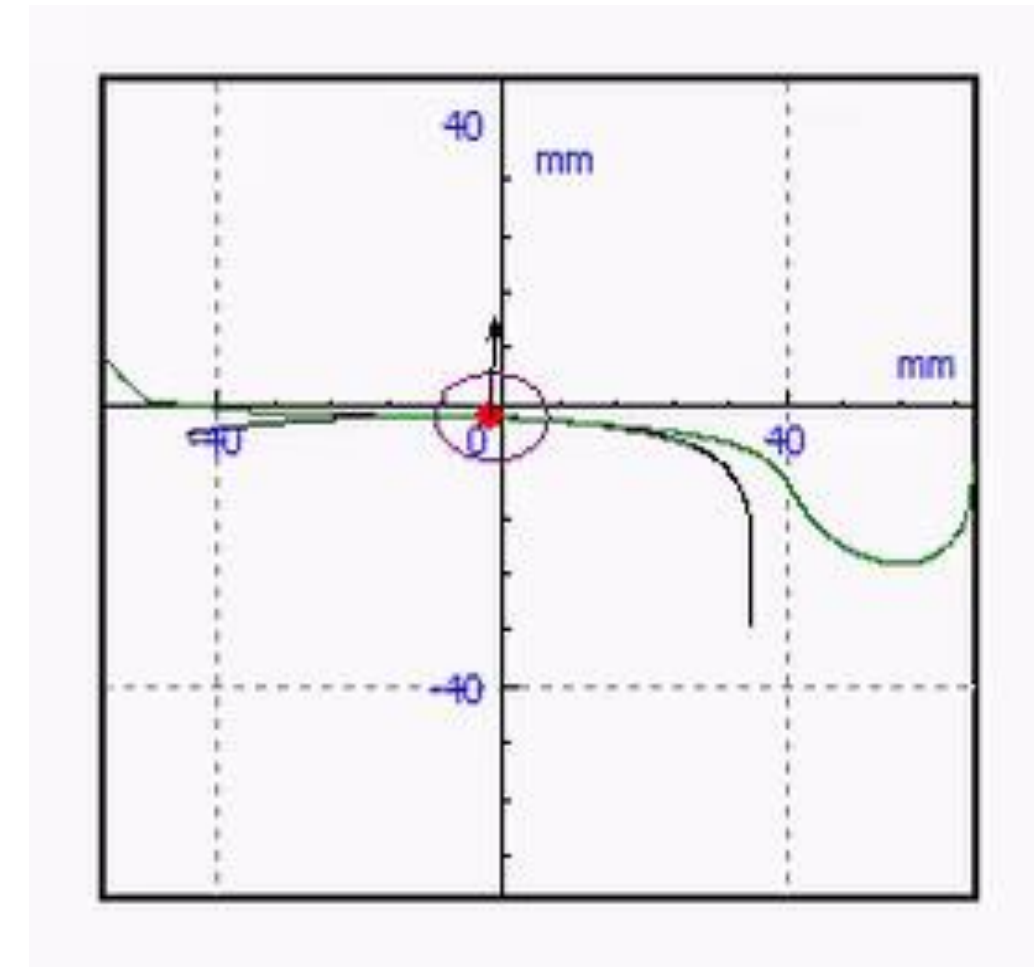
longitudinal
traction/creepage



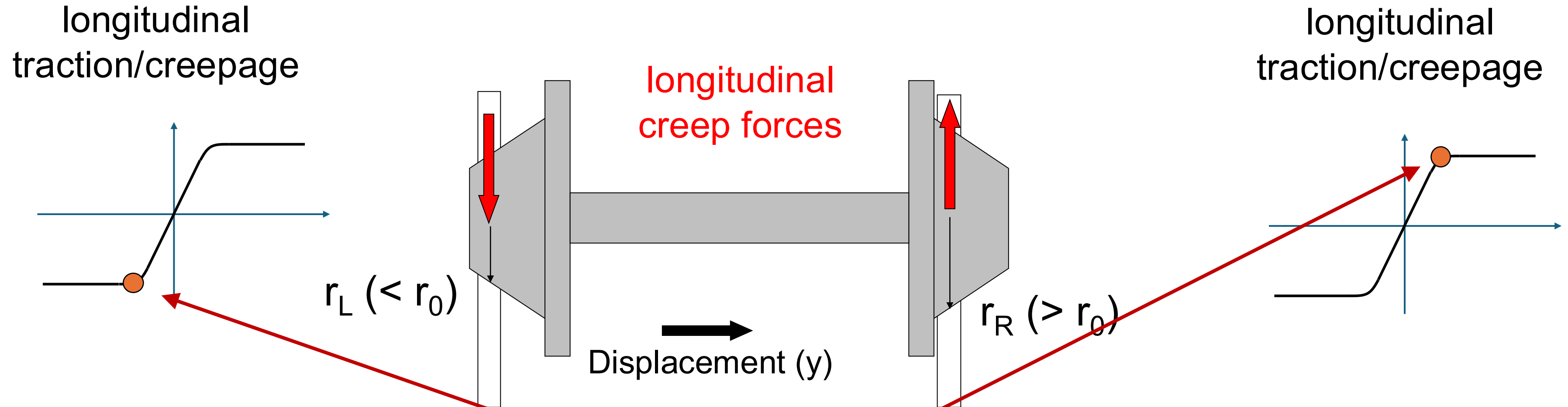


Returning to our opening question:

- As a wheel “climbs” the rail, its radius starts getting much larger, very quickly.
- Why wouldn’t the resulting rolling radius difference between the wheels be enough to steer the wheelset back on track instead of continuing to derail?



First, thinking (only) about *longitudinal* creepage and remembering our wheelset and steering forces...



- What happens once we've reached **this** point on our traction-creepage curves?

Next, remembering that there are *interactions between* creepage types

- Consider the following scenario:

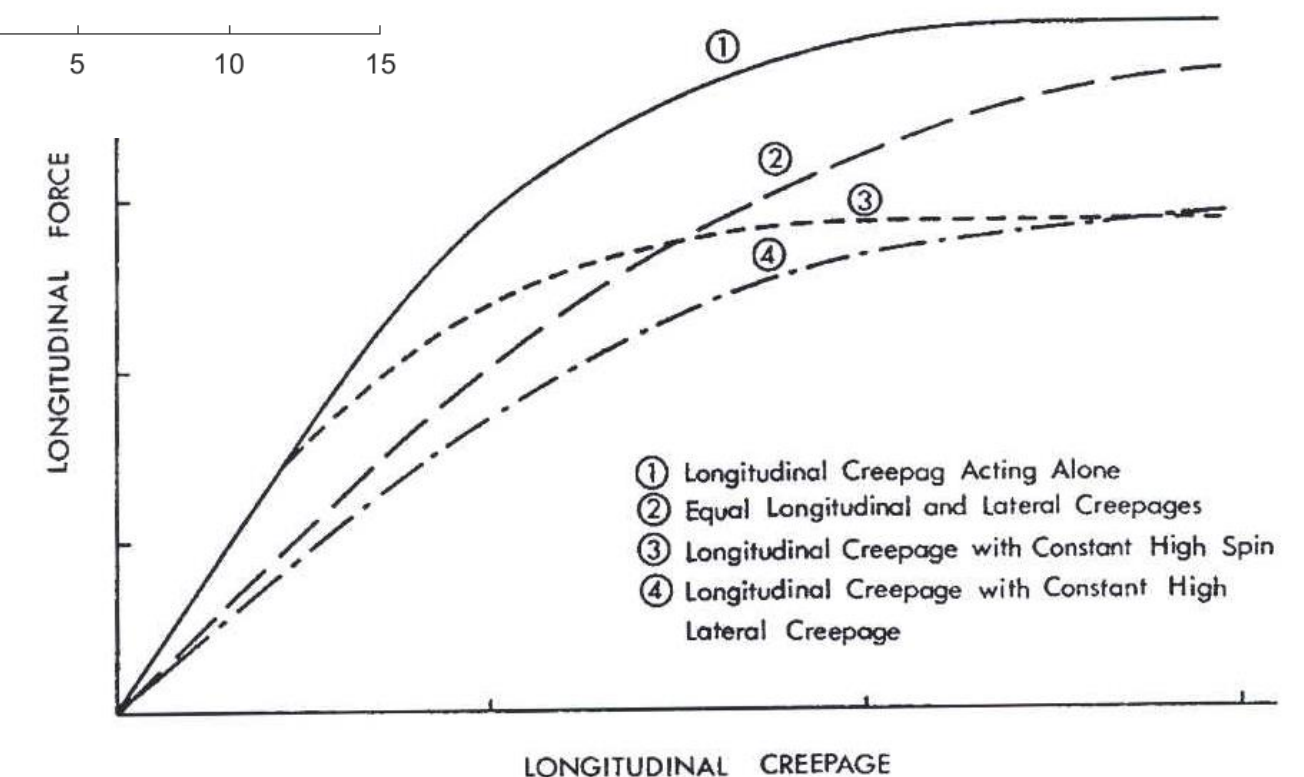
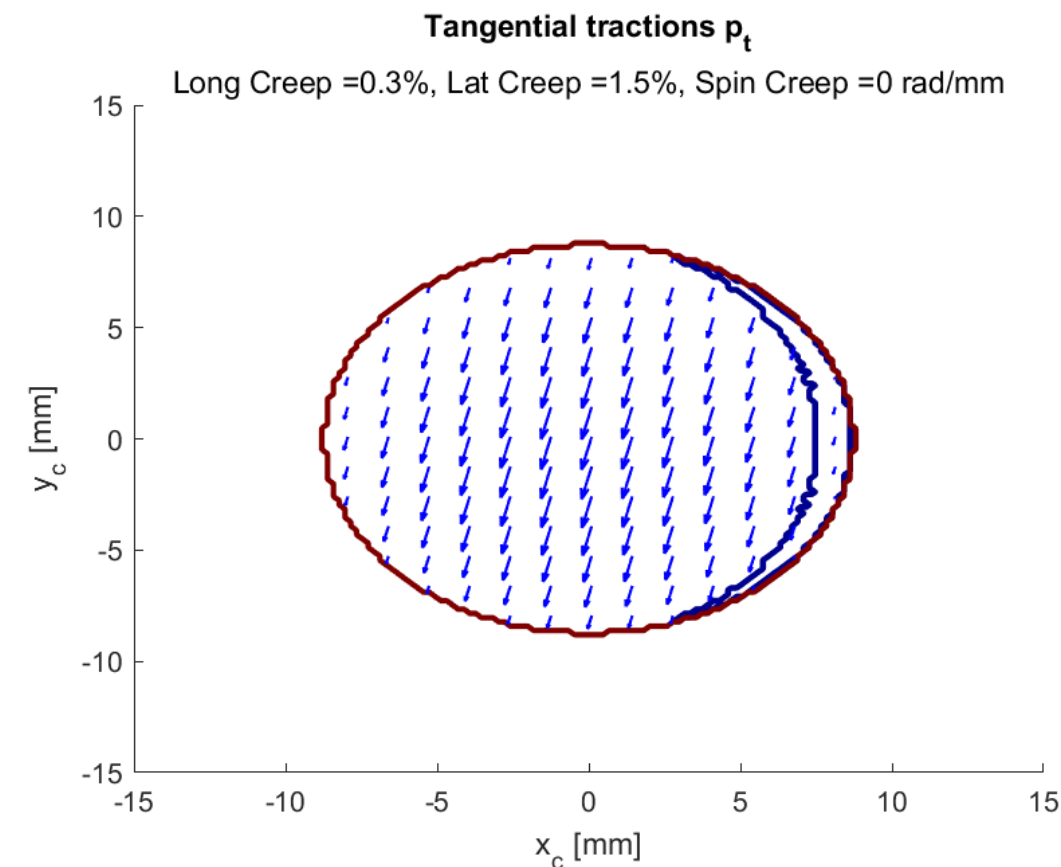
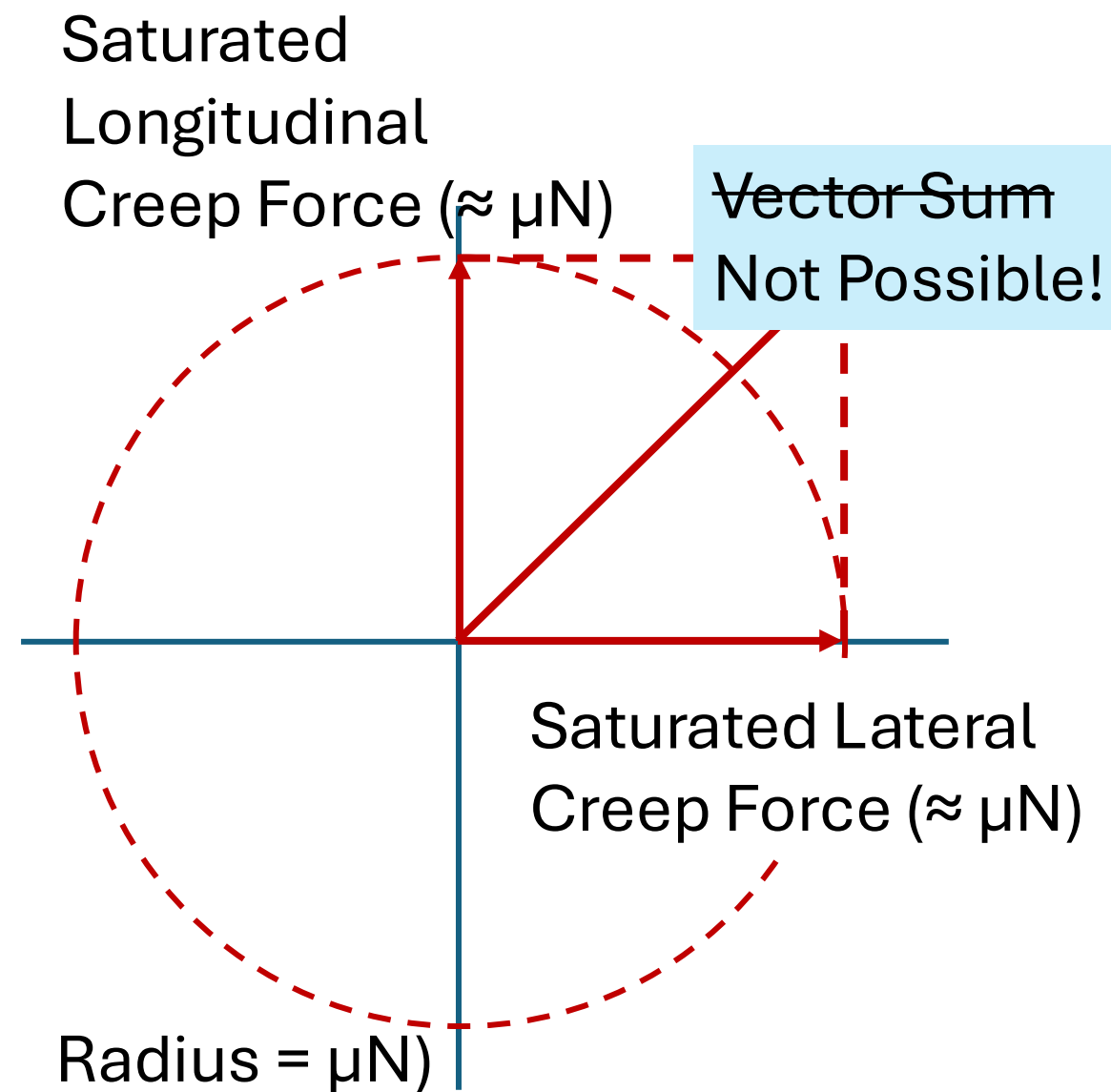
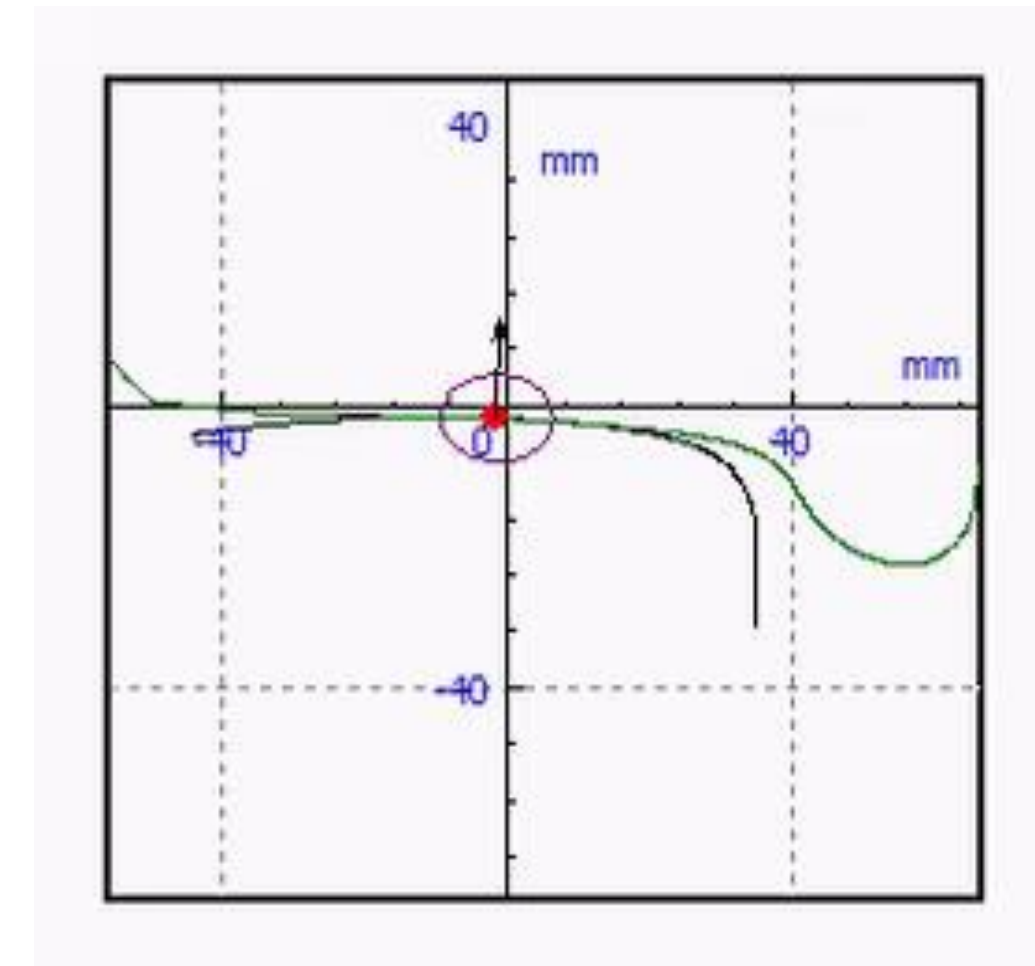


Figure 9: Several examples of longitudinal traction force reduction due to presence of lateral or spin creepages in the interface

Returning to our opening question:

- **At low angles of attack:** once our longitudinal friction forces are saturated, further increases in RRD won't produce significant increases in steering moment (it's basically a fight with gravity).
- **At higher angles of attack:** the presence of lateral creepage will also (sometimes greatly) reduce the amount of longitudinal creep force that is 'available' for steering.
- **Bottom line:** If L/V conditions are severe enough to overcome gravity and get the flange tip onto the top of rail, they are more than enough to keep it moving once it gets there!





Conclusions

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Thank You